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(54) **ABRASIVE ARTICLES AND METHODS OF FORMING SAME**

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(56) **References Cited**

U.S. PATENT DOCUMENTS

345,604 A	7/1886	Semper
1,910,444 A	5/1933	Nicholson
2,033,991 A	3/1936	Melton et al.
2,036,903 A	4/1936	Webster
2,049,874 A	8/1936	Sherk
(Continued)		

FOREIGN PATENT DOCUMENTS

CA	743715 A	10/1966
CA	2423788 A1	7/2002
(Continued)		

OTHER PUBLICATIONS

Torre, "Investigation of Shaped Abrasive Particles vol. 1: Review of
U.S. Pat. No. 6,054,093 Apr. 25, 2000" © Apr. 2011.

(Continued)

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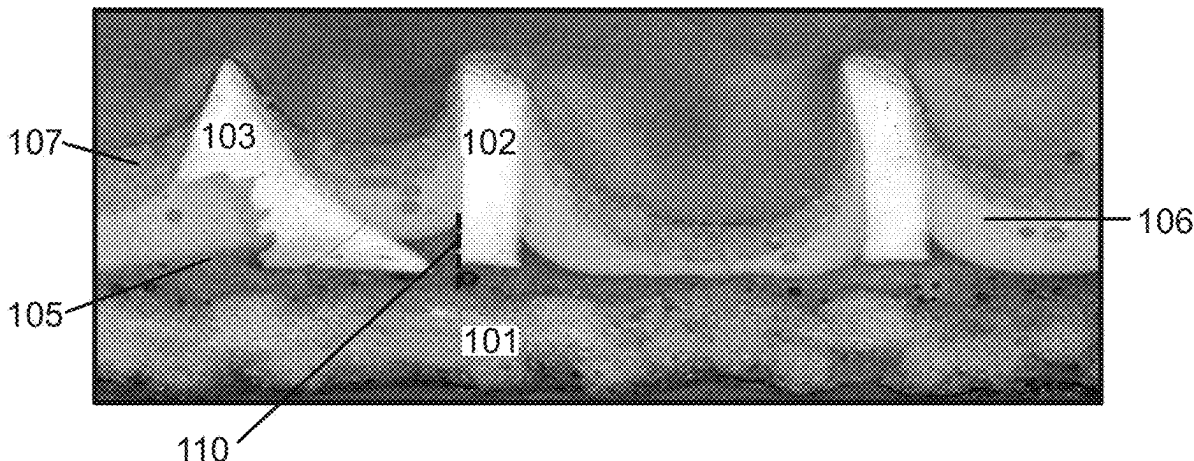
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(57) **ABSTRACT**

An abrasive article includes a backing; a make coat over-
lying the backing; and a plurality of abrasive particles
overlying the backing and at least partially contained in the
make coat; and a make coat thickness ratio (T_g/T_a) of not
greater than 1.5, wherein T_g is the average thickness of the
make coat at the sides of the abrasive particles and the T_a is
the average thickness of the make coat.

28 Claims, 13 Drawing Sheets

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(56)

References Cited

U.S. PATENT DOCUMENTS

2,148,400	A	2/1939	Crompton, Jr.	4,770,671	A	9/1988	Monroe
2,248,064	A	7/1941	Carlton et al.	4,786,292	A	11/1988	Janz et al.
2,248,990	A	7/1941	Heany	4,797,139	A	1/1989	Bauer
2,290,877	A	7/1942	Heany	4,797,269	A	1/1989	Bauer et al.
2,318,360	A	5/1943	Benner et al.	4,799,939	A	1/1989	Bloecher et al.
2,376,343	A	5/1945	Carlton	4,829,027	A	5/1989	Cutler et al.
2,563,650	A	8/1951	Heinemann	4,832,706	A	5/1989	Yates
2,880,080	A	3/1959	Rankin et al.	4,848,041	A	7/1989	Kruschke
3,041,156	A	6/1962	Rowse et al.	4,858,527	A	8/1989	Masanao
3,067,551	A	12/1962	Maginnis	4,863,573	A	9/1989	Moore et al.
3,079,242	A	2/1963	Glasgow	4,876,226	A	10/1989	Fuentes
3,079,243	A	2/1963	Ueltz	4,881,951	A	11/1989	Wood et al.
3,123,948	A	3/1964	Kistler et al.	4,917,852	A	4/1990	Poole et al.
3,141,271	A	7/1964	Fischer et al.	4,918,116	A	4/1990	Gardziella et al.
3,276,852	A	10/1966	Lemelson	4,925,457	A	5/1990	Dekok et al.
3,377,660	A	4/1968	Marshall et al.	4,925,815	A	5/1990	Tani et al.
3,379,543	A	4/1968	Norwalk	4,930,266	A	6/1990	Calhoun et al.
3,387,957	A	6/1968	Howard	4,942,011	A	7/1990	Bolt et al.
3,454,385	A	7/1969	Amero	4,954,462	A	9/1990	Wood
3,477,180	A	11/1969	Robertson, Jr. et al.	4,960,441	A	10/1990	Pellow et al.
3,480,395	A	11/1969	McMullen et al.	4,961,757	A	10/1990	Rhodes et al.
3,481,723	A	12/1969	Kistler et al.	4,963,012	A	10/1990	Tracy
3,491,492	A	1/1970	Ueltz	4,964,883	A	10/1990	Morris et al.
3,495,359	A	2/1970	Smith et al.	4,970,057	A	11/1990	Wilkins et al.
3,536,005	A	10/1970	Derrickson	4,997,461	A	3/1991	Markhoff-Matheny et al.
3,590,799	A	7/1971	Guuchowicz	5,000,760	A	3/1991	Ohtsubo et al.
3,608,050	A	9/1971	Carman et al.	5,008,222	A	4/1991	Kameda
3,608,134	A	9/1971	Cook	5,009,675	A	4/1991	Kunz et al.
3,615,308	A	10/1971	Amero	5,009,676	A	4/1991	Rue et al.
3,619,151	A	11/1971	Sheets, Jr. et al.	5,011,508	A	4/1991	Wald et al.
3,637,360	A	1/1972	Ueltz	5,011,510	A	4/1991	Hayakawa et al.
3,670,467	A	6/1972	Walker	5,014,468	A	5/1991	Ravipati et al.
3,672,934	A	6/1972	Larry	5,024,795	A	6/1991	Kennedy et al.
3,808,747	A	5/1974	Kenagy	5,032,304	A	7/1991	Toyota
3,819,785	A	6/1974	Argyle et al.	5,035,723	A	7/1991	Kalinowski et al.
3,859,407	A	1/1975	Blanding et al.	5,035,724	A	7/1991	Pukari et al.
3,874,856	A	4/1975	Leeds	5,042,991	A	8/1991	Kunz et al.
3,909,991	A	10/1975	Coes, Jr.	5,049,165	A	9/1991	Tselesin
3,940,276	A	2/1976	Wilson	5,049,166	A	9/1991	Kirkendall
3,950,148	A	4/1976	Fukuda et al.	5,049,645	A	9/1991	Nagaoka et al.
3,960,577	A	6/1976	Prochazka	5,053,367	A	10/1991	Newkirk et al.
3,977,132	A	8/1976	Sekigawa	5,053,369	A	10/1991	Winkler et al.
3,986,885	A	10/1976	Lankard	5,076,991	A	12/1991	Poole et al.
3,991,527	A	11/1976	Maran	5,078,753	A	1/1992	Broberg et al.
4,004,934	A	1/1977	Prochazka	5,081,082	A	1/1992	Hai-Doo et al.
4,037,367	A	7/1977	Kruse	5,085,671	A	2/1992	Martin et al.
4,045,919	A	9/1977	Moritomo	5,090,968	A	2/1992	Pellow
4,055,451	A	10/1977	Cockbain et al.	5,094,986	A	3/1992	Matsumoto et al.
4,073,096	A	2/1978	Ueltz et al.	5,098,740	A	3/1992	Tewari
4,114,322	A	9/1978	Greenspan	5,103,598	A	4/1992	Kelly
4,150,078	A	4/1979	Miller et al.	5,108,963	A	4/1992	Fu et al.
4,194,887	A	3/1980	Ueltz et al.	5,114,438	A	5/1992	Leatherman et al.
4,252,544	A	2/1981	Takahashi	5,120,327	A	6/1992	Dennis
4,261,706	A	4/1981	Blanding et al.	5,123,935	A	6/1992	Kanamaru et al.
4,286,905	A	9/1981	Samanta	5,129,919	A	7/1992	Kalinowski et al.
4,304,576	A	12/1981	Hattori et al.	5,131,926	A	7/1992	Rostoker et al.
4,314,827	A	2/1982	Leitheiser et al.	5,132,984	A	7/1992	Simpson
4,341,663	A	7/1982	Derleth et al.	5,139,978	A	8/1992	Wood
4,393,021	A	7/1983	Eisenberg et al.	5,152,917	A	10/1992	Pieper et al.
4,452,911	A	6/1984	Eccles et al.	5,160,509	A	11/1992	Carman et al.
4,457,767	A	7/1984	Poon et al.	5,164,744	A	11/1992	Yoshida et al.
4,469,758	A	9/1984	Scott	5,173,457	A	12/1992	Shorthouse
4,505,720	A	3/1985	Gabor et al.	5,178,849	A	1/1993	Bauer
4,541,842	A	9/1985	Rostoker	5,180,630	A	1/1993	Giglia
4,548,617	A	10/1985	Miyatani et al.	5,185,012	A	2/1993	Kelly
4,570,048	A	2/1986	Poole	5,185,299	A	2/1993	Wood et al.
4,618,349	A	10/1986	Hashimoto et al.	5,190,568	A	3/1993	Tselesin
4,623,364	A	11/1986	Cottringer et al.	5,194,072	A	3/1993	Rue et al.
4,656,330	A	4/1987	Poole	5,201,916	A	4/1993	Berg et al.
4,657,754	A	4/1987	Bauer et al.	5,203,886	A	4/1993	Sheldon et al.
4,659,341	A	4/1987	Ludwig et al.	5,213,591	A	5/1993	Celikkaya et al.
4,678,560	A	7/1987	Stole et al.	5,215,552	A	6/1993	Sung
4,711,750	A	12/1987	Scott	5,219,462	A	6/1993	Bruxvoort et al.
4,728,043	A	3/1988	Ersdal et al.	5,219,806	A	6/1993	Wood
4,744,802	A	5/1988	Schwabel	5,221,294	A	6/1993	Carman et al.
				5,224,970	A	7/1993	Harakawa et al.
				5,227,104	A	7/1993	Bauer
				5,244,477	A	9/1993	Rue et al.
				5,244,849	A	9/1993	Roy et al.

(56)

References Cited

U.S. PATENT DOCUMENTS

5,273,558 A	12/1993	Nelson et al.	5,725,162 A	3/1998	Garg et al.
5,277,702 A	1/1994	Thibault et al.	5,736,619 A	4/1998	Kane et al.
5,282,875 A	2/1994	Wood	5,738,696 A	4/1998	Wu
5,288,297 A	2/1994	Ringwood	5,738,697 A	4/1998	Wu et al.
5,300,130 A	4/1994	Rostoker	5,751,313 A	5/1998	Miyashita et al.
5,304,331 A	4/1994	Leonard et al.	5,759,481 A	6/1998	Pujari et al.
5,312,789 A	5/1994	Wood	5,776,214 A	7/1998	Wood
5,312,791 A	5/1994	Coblentz et al.	5,779,743 A	7/1998	Wood
5,314,513 A	5/1994	Miller et al.	5,785,722 A	7/1998	Garg et al.
5,366,523 A	11/1994	Rowenhorst et al.	5,810,587 A	9/1998	Bruns et al.
5,366,525 A	11/1994	Fujiyama	5,820,450 A	10/1998	Calhoun
5,372,620 A	12/1994	Rowse et al.	5,830,248 A	11/1998	Christianson et al.
5,373,786 A	12/1994	Umaba	5,840,089 A	11/1998	Chesley et al.
5,376,598 A	12/1994	Preedy et al.	5,849,646 A	12/1998	Stout et al.
5,376,602 A	12/1994	Nilsen	5,855,997 A	1/1999	Amateau
5,383,945 A	1/1995	Cottringer et al.	5,863,306 A	1/1999	Wei et al.
5,395,407 A	3/1995	Cottringer et al.	5,866,254 A	2/1999	Peker et al.
5,409,645 A	4/1995	Torre, Jr. et al.	5,876,793 A	3/1999	Sherman et al.
5,429,648 A	7/1995	Wu	5,885,311 A	3/1999	McCutcheon et al.
5,431,967 A	7/1995	Manthiram	5,893,935 A	4/1999	Wood
5,435,816 A	7/1995	Spurgeon et al.	5,902,647 A	5/1999	Venkataramani
5,437,754 A	8/1995	Calhoun	5,908,477 A	6/1999	Harmer et al.
5,441,549 A	8/1995	Helmin	5,908,478 A	6/1999	Wood
5,443,603 A	8/1995	Kirkendall	5,919,549 A	7/1999	Van et al.
5,447,894 A	9/1995	Yasuoka et al.	5,924,917 A	7/1999	Benedict et al.
5,453,106 A	9/1995	Roberts	5,946,991 A	9/1999	Hoopman
5,454,844 A	10/1995	Hibbard et al.	5,975,987 A	11/1999	Hoopman et al.
5,470,806 A	11/1995	Krstic et al.	5,980,678 A	11/1999	Tselesin
5,479,873 A	1/1996	Shintani et al.	5,984,988 A	11/1999	Berg et al.
5,482,756 A	1/1996	Berger et al.	5,989,301 A	11/1999	Laconto, Sr. et al.
5,486,496 A	1/1996	Talbert et al.	5,997,597 A	12/1999	Hagan
5,489,318 A	2/1996	Erickson et al.	6,016,660 A	1/2000	Abramshe
5,496,386 A	3/1996	Broberg et al.	6,019,805 A	2/2000	Herron
5,498,268 A	3/1996	Gagliardi et al.	6,024,824 A	2/2000	Krech
5,500,273 A	3/1996	Holmes et al.	6,027,326 A	2/2000	Cesarano, III et al.
5,514,631 A	5/1996	Cottringer et al.	6,039,775 A	3/2000	Ho et al.
5,516,347 A	5/1996	Garg	6,048,577 A	4/2000	Garg
5,516,348 A	5/1996	Conwell et al.	6,053,956 A	4/2000	Wood
5,523,074 A	6/1996	Takahashi et al.	6,054,093 A	4/2000	Torre, Jr. et al.
5,525,100 A	6/1996	Kelly et al.	6,080,215 A	6/2000	Stubbs et al.
5,527,369 A	6/1996	Garg	6,080,216 A	6/2000	Erickson
5,543,368 A	8/1996	Talbert et al.	6,083,622 A	7/2000	Garg et al.
5,549,962 A	8/1996	Holmes et al.	6,096,107 A	8/2000	Caracostas et al.
5,551,963 A	9/1996	Larmie	6,110,241 A	8/2000	Sung
5,560,745 A	10/1996	Roberts	6,129,540 A	10/2000	Hoopman et al.
5,567,150 A	10/1996	Conwell et al.	6,136,288 A	10/2000	Bauer et al.
5,567,214 A	10/1996	Ashley	6,146,247 A	11/2000	Nokubi et al.
5,567,251 A	10/1996	Peker et al.	6,179,887 B1	1/2001	Barber, Jr. et al.
5,571,297 A	11/1996	Swei et al.	6,206,942 B1	3/2001	Wood
5,576,409 A	11/1996	Mackey	6,228,134 B1	5/2001	Erickson
5,578,095 A	11/1996	Bland et al.	6,238,450 B1	5/2001	Garg et al.
5,578,222 A	11/1996	Trischuk et al.	6,258,137 B1	7/2001	Garg et al.
5,582,625 A	12/1996	Wright et al.	6,258,141 B1	7/2001	Sung et al.
5,584,896 A	12/1996	Broberg et al.	6,261,682 B1	7/2001	Law
5,584,897 A	12/1996	Christianson et al.	6,264,710 B1	7/2001	Erickson
5,591,685 A	1/1997	Mitomo et al.	6,277,160 B1	8/2001	Stubbs et al.
5,593,468 A	1/1997	Khaund et al.	6,277,161 B1	8/2001	Castro et al.
5,599,493 A	2/1997	Ito et al.	6,283,997 B1	9/2001	Garg et al.
5,603,738 A	2/1997	Zeiringer et al.	6,284,690 B1	9/2001	Nakahata et al.
5,609,706 A	3/1997	Benedict et al.	6,287,353 B1	9/2001	Celikkaya
5,611,829 A	3/1997	Monroe et al.	6,306,007 B1	10/2001	Mori et al.
5,618,221 A	4/1997	Furukawa et al.	6,312,324 B1	11/2001	Mitsui et al.
5,628,952 A	5/1997	Holmes et al.	6,319,108 B1	11/2001	Adefris et al.
5,641,469 A	6/1997	Garg et al.	6,331,343 B1	12/2001	Perez et al.
RE35,570 E	7/1997	Rowenhorst et al.	6,371,842 B1	4/2002	Romero
5,645,619 A	7/1997	Erickson et al.	6,391,812 B1	5/2002	Araki et al.
5,651,925 A	7/1997	Ashley et al.	6,398,989 B1	6/2002	Bergstrom
5,656,217 A	8/1997	Rogers et al.	6,401,795 B1	6/2002	Cesarano, III et al.
5,667,542 A	9/1997	Law et al.	6,403,001 B1	6/2002	Hayashi
5,669,941 A	9/1997	Peterson	6,406,200 B2	6/2002	Mahoney
5,669,943 A	9/1997	Horton et al.	6,413,286 B1	7/2002	Swei et al.
5,672,097 A	9/1997	Hoopman	6,428,392 B1	8/2002	Sunahara et al.
5,672,554 A	9/1997	Mohri et al.	6,451,076 B1	9/2002	Nevoret et al.
5,683,844 A	11/1997	Mammino	6,475,253 B2	11/2002	Culler et al.
5,702,811 A	12/1997	Ho et al.	6,500,493 B2	12/2002	Swei et al.
			6,511,938 B1	1/2003	Liu
			6,524,681 B1	2/2003	Seitz et al.
			6,531,423 B1	3/2003	Schwetz et al.
			6,537,140 B1	3/2003	Miller et al.

(56)

References Cited

U.S. PATENT DOCUMENTS

6,579,819	B2	6/2003	Hirosaki et al.	8,251,774	B2	8/2012	Joseph et al.
6,582,623	B1	6/2003	Grumbine et al.	8,256,091	B2	9/2012	Duescher
6,583,080	B1	6/2003	Rosenflanz	8,333,360	B2	12/2012	Rule et al.
6,599,177	B2	7/2003	Nevoret et al.	8,440,602	B2	5/2013	Gonzales et al.
6,620,214	B2	9/2003	McArdle	8,440,603	B2	5/2013	Gonzales et al.
6,646,019	B2	11/2003	Perez et al.	8,445,422	B2	5/2013	Gonzales et al.
6,652,361	B1	11/2003	Gash et al.	8,470,759	B2	6/2013	Gonzales et al.
6,669,745	B2	12/2003	Prichard et al.	8,480,772	B2	7/2013	Welygan et al.
6,685,755	B2	2/2004	Ramanath et al.	8,530,682	B2	9/2013	Sachs
6,696,258	B1	2/2004	Wei et al.	8,568,497	B2	10/2013	Sheridan
6,702,650	B2	3/2004	Adefris	8,628,597	B2	1/2014	Palmgren et al.
6,737,378	B2	5/2004	Hirosaki et al.	8,783,589	B2	7/2014	Hart et al.
6,749,496	B2	6/2004	Mota et al.	8,852,643	B2	10/2014	Gonzales et al.
6,750,173	B2	6/2004	Rizkalla	8,920,527	B2	12/2014	Seider et al.
6,752,700	B2	6/2004	Duescher	8,921,687	B1	12/2014	Welser et al.
6,755,729	B2	6/2004	Ramanath et al.	9,017,439	B2	4/2015	Yener et al.
6,802,878	B1	10/2004	Monroe	9,079,154	B2	7/2015	Rosendahl et al.
6,821,196	B2	11/2004	Oliver	9,181,477	B2	11/2015	Collins et al.
6,833,014	B2	12/2004	Welygan et al.	9,211,634	B2	12/2015	Rehrig et al.
6,843,815	B1	1/2005	Thurber et al.	9,259,726	B2	2/2016	Gopal
6,846,795	B2	1/2005	Lant et al.	9,375,826	B2	6/2016	Tian et al.
6,878,456	B2	4/2005	Castro et al.	9,717,674	B1	8/2017	Guskey et al.
6,881,483	B2	4/2005	McArdle et al.	9,758,724	B2	9/2017	Collins et al.
6,888,360	B1	5/2005	Connell et al.	9,982,175	B2	5/2018	Sarangi et al.
6,913,824	B2	7/2005	Culler et al.	D849,066	S	5/2019	Hanschen et al.
6,942,561	B2	9/2005	Mota et al.	D849,067	S	5/2019	Hanschen et al.
6,949,128	B2	9/2005	Annen	10,351,745	B2	7/2019	Josseaux et al.
6,951,504	B2	10/2005	Adefris et al.	10,364,383	B2	7/2019	Yener et al.
6,974,930	B2	12/2005	Jense	D862,538	S	10/2019	Hanschen et al.
7,022,179	B1	4/2006	Dry	D870,782	S	12/2019	Hanschen et al.
7,044,989	B2	5/2006	Welygan et al.	10,556,323	B2	2/2020	Alkhas et al.
7,112,621	B2	9/2006	Rohrbaugh et al.	10,557,068	B2	2/2020	Oldenkotte et al.
7,141,522	B2	11/2006	Rosenflanz et al.	10,563,105	B2	2/2020	Cotter et al.
7,168,267	B2	1/2007	Rosenflanz et al.	10,655,038	B2	5/2020	Martinez et al.
7,169,198	B2	1/2007	Moeltgen et al.	10,710,211	B2	7/2020	Lehuu et al.
7,267,604	B2	9/2007	Yoshizawa et al.	10,717,908	B2	7/2020	Hejtmann et al.
7,267,700	B2	9/2007	Collins et al.	2001/0027623	A1	10/2001	Rosenflanz
7,294,158	B2	11/2007	Welygan et al.	2002/0026752	A1	3/2002	Culler et al.
7,297,170	B2	11/2007	Welygan et al.	2002/0068518	A1	6/2002	Cesena et al.
7,297,402	B2	11/2007	Evans et al.	2002/0084290	A1	7/2002	Materna
7,364,788	B2	4/2008	Kishbaugh et al.	2002/0090891	A1	7/2002	Adefris et al.
7,373,887	B2	5/2008	Jackson	2002/0151265	A1	10/2002	Adefris
7,384,437	B2	6/2008	Welygan et al.	2002/0170236	A1	11/2002	Larson et al.
7,404,832	B2	7/2008	Ohtsubo et al.	2002/0174935	A1	11/2002	Burdon et al.
7,488,544	B2	2/2009	Schofalvi et al.	2002/0177391	A1	11/2002	Fritz et al.
7,507,268	B2	3/2009	Rosenflanz	2003/0008933	A1	1/2003	Perez et al.
7,553,346	B2	6/2009	Welygan et al.	2003/0022961	A1	1/2003	Kusaka et al.
7,556,558	B2	7/2009	Palmgren	2003/0029094	A1	2/2003	Moeltgen et al.
7,560,062	B2	7/2009	Gould et al.	2003/0085204	A1	5/2003	Lagos
7,560,139	B2	7/2009	Thebault et al.	2003/0109371	A1	6/2003	Pujari et al.
7,563,293	B2	7/2009	Rosenflanz	2003/0110707	A1	6/2003	Rosenflanz et al.
7,611,795	B2	11/2009	Aoyama et al.	2003/0126800	A1	7/2003	Seth et al.
7,618,684	B2	11/2009	Nesbitt	2003/0228738	A1	12/2003	Beaudoin
7,632,434	B2	12/2009	Duescher	2004/0003895	A1	1/2004	Amano et al.
7,651,386	B2	1/2010	Sung	2004/0148868	A1	8/2004	Anderson et al.
7,662,735	B2	2/2010	Rosenflanz et al.	2004/0148967	A1	8/2004	Celikaya et al.
7,666,344	B2	2/2010	Schofalvi et al.	2004/0202844	A1	10/2004	Wong
7,666,475	B2	2/2010	Morrison	2004/0224125	A1	11/2004	Yamada et al.
7,669,658	B2	3/2010	Barron et al.	2004/0235406	A1	11/2004	Duescher
7,670,679	B2	3/2010	Krishna et al.	2004/0244675	A1	12/2004	Kishimoto et al.
7,695,542	B2	4/2010	Drivdahl et al.	2005/0020190	A1	1/2005	Schutz et al.
7,858,189	B2	12/2010	Wagener et al.	2005/0060941	A1	3/2005	Provow et al.
7,867,302	B2	1/2011	Nevoret et al.	2005/0060947	A1	3/2005	McArdle et al.
7,906,057	B2	3/2011	Zhang et al.	2005/0064805	A1	3/2005	Culler et al.
7,968,147	B2	6/2011	Fang et al.	2005/0081455	A1	4/2005	Welygan et al.
7,972,430	B2	7/2011	Millard et al.	2005/0118939	A1	6/2005	Duescher
8,021,449	B2	9/2011	Seth et al.	2005/0132655	A1	6/2005	Anderson et al.
8,034,137	B2	10/2011	Erickson et al.	2005/0218565	A1	10/2005	DiChiara, Jr.
8,049,136	B2	11/2011	Mase et al.	2005/0223649	A1	10/2005	O'Gary et al.
8,070,556	B2	12/2011	Kumar et al.	2005/0232853	A1	10/2005	Evans et al.
8,123,828	B2	2/2012	Culler et al.	2005/0245179	A1	11/2005	Luedeke
8,141,484	B2	3/2012	Ojima et al.	2005/0255801	A1	11/2005	Pollasky
8,142,531	B2	3/2012	Adefris et al.	2005/0266221	A1	12/2005	Karam et al.
8,142,532	B2	3/2012	Erickson et al.	2005/0271795	A1	12/2005	Moini et al.
8,142,891	B2	3/2012	Culler et al.	2005/0284029	A1	12/2005	Bourlier et al.
				2006/0049540	A1	3/2006	Hui et al.
				2006/0126265	A1	6/2006	Crespi et al.
				2006/0135050	A1	6/2006	Petersen et al.
				2006/0177488	A1	8/2006	Caruso et al.

(56)

References Cited

U.S. PATENT DOCUMENTS

2006/0185256 A1	8/2006	Nevoret et al.	2013/0000212 A1	1/2013	Wang et al.
2007/0011951 A1	1/2007	Gaeta et al.	2013/0000216 A1	1/2013	Wang et al.
2007/0020457 A1	1/2007	Adefris	2013/0009484 A1	1/2013	Yu
2007/0051355 A1	3/2007	Sung	2013/0036402 A1	2/2013	Mutisya et al.
2007/0072527 A1	3/2007	Palmgren	2013/0045251 A1	2/2013	Cen et al.
2007/0074456 A1	4/2007	Orlhac et al.	2013/0067669 A1	3/2013	Gonzales et al.
2007/0087928 A1	4/2007	Rosenflanz et al.	2013/0072417 A1	3/2013	Perez-Prat et al.
2007/0234646 A1	10/2007	Can et al.	2013/0074418 A1	3/2013	Panzarella et al.
2008/0017053 A1	1/2008	Araumi et al.	2013/0125477 A1	5/2013	Adefris
2008/0072500 A1	3/2008	Klett et al.	2013/0180180 A1	7/2013	Yener et al.
2008/0098659 A1	5/2008	Sung	2013/0186005 A1	7/2013	Kavanaugh
2008/0121124 A1	5/2008	Sato	2013/0186006 A1	7/2013	Kavanaugh et al.
2008/0172951 A1	7/2008	Starling	2013/0199105 A1	8/2013	Braun et al.
2008/0176075 A1	7/2008	Bauer et al.	2013/0203328 A1	8/2013	Givot et al.
2008/0179783 A1	7/2008	Liu et al.	2013/0212952 A1	8/2013	Welygan et al.
2008/0230951 A1	9/2008	Dannoux et al.	2013/0236725 A1	9/2013	Yener et al.
2008/0233845 A1	9/2008	Annen et al.	2013/0255162 A1	10/2013	Welygan et al.
2008/0262577 A1	10/2008	Altshuler et al.	2013/0260656 A1	10/2013	Seth et al.
2008/0271384 A1	11/2008	Puthanangady et al.	2013/0267150 A1	10/2013	Seider et al.
2008/0286590 A1	11/2008	Besida et al.	2013/0283705 A1	10/2013	Fischer et al.
2008/0299875 A1	12/2008	Duescher	2013/0296587 A1	11/2013	Rosendahl
2009/0016916 A1	1/2009	Rosenzweig et al.	2013/0305614 A1	11/2013	Gaeta et al.
2009/0017276 A1	1/2009	Hoglund et al.	2013/0337262 A1	12/2013	Bauer et al.
2009/0017736 A1	1/2009	Block et al.	2013/0337725 A1	12/2013	Monroe
2009/0098365 A1	4/2009	Moeltgen	2013/0344786 A1	12/2013	Keipert
2009/0165394 A1	7/2009	Culler et al.	2014/0000176 A1	1/2014	Moren et al.
2009/0165661 A1	7/2009	Koenig et al.	2014/0007518 A1	1/2014	Yener et al.
2009/0169816 A1	7/2009	Erickson et al.	2014/0080393 A1	3/2014	Ludwig
2009/0208734 A1	8/2009	Macfie et al.	2014/0106126 A1	4/2014	Gaeta et al.
2009/0246464 A1	10/2009	Watanabe et al.	2014/0107356 A1	4/2014	Gopal
2010/0000159 A1	1/2010	Walia et al.	2014/0182216 A1	7/2014	Panzarella et al.
2010/0003900 A1	1/2010	Sakaguchi et al.	2014/0182217 A1	7/2014	Yener et al.
2010/0003904 A1	1/2010	Duescher	2014/0186585 A1	7/2014	Field, III et al.
2010/0040767 A1	2/2010	Uibel et al.	2014/0250797 A1	9/2014	Yener et al.
2010/0056816 A1	3/2010	Wallin et al.	2014/0256238 A1	9/2014	Van et al.
2010/0064594 A1	3/2010	Pakalapati et al.	2014/0287658 A1	9/2014	Flaschberger et al.
2010/0068974 A1	3/2010	Dumm	2014/0290147 A1	10/2014	Seth et al.
2010/0146867 A1	6/2010	Boden et al.	2014/0325917 A1	11/2014	Czerepinski et al.
2010/0151195 A1	6/2010	Culler et al.	2014/0345204 A1	11/2014	Wang et al.
2010/0151196 A1	6/2010	Adefris et al.	2014/0345205 A1	11/2014	Kavanaugh et al.
2010/0151201 A1	6/2010	Erickson et al.	2014/0352721 A1	12/2014	Gonzales et al.
2010/0190424 A1	7/2010	Francois et al.	2014/0352722 A1	12/2014	Gonzales et al.
2010/0201018 A1	8/2010	Yoshioka et al.	2014/0357544 A1	12/2014	Gonzales et al.
2010/0251625 A1	10/2010	Gaeta	2014/0378036 A1	12/2014	Cichowlas et al.
2010/0292428 A1	11/2010	Meador et al.	2015/0000209 A1	1/2015	Louapre et al.
2010/0307067 A1	12/2010	Sigalas et al.	2015/0000210 A1	1/2015	Breder et al.
2010/0319269 A1	12/2010	Erickson	2015/0007399 A1	1/2015	Gonzales et al.
2010/0330886 A1	12/2010	Wu et al.	2015/0007400 A1	1/2015	Gonzales et al.
2011/0008604 A1	1/2011	Boylan	2015/0068130 A1	3/2015	Louapre et al.
2011/0081848 A1	4/2011	Chen	2015/0089881 A1	4/2015	Stevenson et al.
2011/0092137 A1	4/2011	Ohishi et al.	2015/0126098 A1	5/2015	Eilers et al.
2011/0111563 A1	5/2011	Yanagi et al.	2015/0128505 A1	5/2015	Wang et al.
2011/0124483 A1	5/2011	Shah et al.	2015/0183089 A1	7/2015	Iyengar et al.
2011/0136659 A1	6/2011	Allen et al.	2015/0209932 A1	7/2015	Lehuu et al.
2011/0146509 A1	6/2011	Welygan et al.	2015/0218430 A1	8/2015	Yener et al.
2011/0152548 A1	6/2011	Sachs	2015/0232727 A1	8/2015	Erickson
2011/0160104 A1	6/2011	Wu et al.	2015/0267099 A1	9/2015	Panzarella et al.
2011/0244769 A1	10/2011	David et al.	2015/0291865 A1	10/2015	Breder et al.
2011/0289854 A1	12/2011	Moren et al.	2015/0291866 A1	10/2015	Arcona et al.
2011/0314746 A1	12/2011	Erickson et al.	2015/0291867 A1	10/2015	Breder et al.
2012/0000135 A1	1/2012	Eilers et al.	2015/0343603 A1	12/2015	Breder et al.
2012/0034847 A1	2/2012	Besse et al.	2016/0053151 A1	2/2016	Bauer et al.
2012/0055098 A1	3/2012	Ramanath et al.	2016/0090516 A1	3/2016	Yener et al.
2012/0100366 A1	4/2012	Dumm et al.	2016/0107290 A1	4/2016	Bajaj et al.
2012/0137597 A1	6/2012	Adefris et al.	2016/0177152 A1	6/2016	Braun
2012/0142259 A1	6/2012	Hamilton	2016/0177153 A1	6/2016	Josseaux
2012/0144754 A1	6/2012	Culler et al.	2016/0177154 A1	6/2016	Josseaux et al.
2012/0144755 A1	6/2012	Erickson et al.	2016/0186028 A1	6/2016	Louapre et al.
2012/0153547 A1	6/2012	Bauer et al.	2016/0214903 A1	7/2016	Humpal et al.
2012/0167481 A1	7/2012	Yener et al.	2016/0289520 A1	10/2016	Bujnowski et al.
2012/0168979 A1	7/2012	Bauer et al.	2016/0289521 A1	10/2016	Colet et al.
2012/0227333 A1	9/2012	Adefris et al.	2016/0298013 A1	10/2016	Bock et al.
2012/0231711 A1	9/2012	Keipert et al.	2016/0303704 A1	10/2016	Chou et al.
2012/0308837 A1	12/2012	Schlechtriemen et al.	2016/0303705 A1	10/2016	Chou et al.
2012/0321567 A1	12/2012	Gonzales et al.	2016/0304760 A1	10/2016	Bock et al.
			2016/0311081 A1	10/2016	Culler et al.
			2016/0311084 A1	10/2016	Culler et al.
			2016/0326416 A1	11/2016	Bauer et al.
			2016/0340564 A1	11/2016	Louapre et al.

(56)

References Cited

U.S. PATENT DOCUMENTS

2016/0354898 A1 12/2016 Nienaber et al.
 2016/0362589 A1 12/2016 Bauer et al.
 2016/0375556 A1 12/2016 Seth et al.
 2017/0015886 A1 1/2017 Czerepinski et al.
 2017/0028531 A1 2/2017 Gaeta et al.
 2017/0050293 A1 2/2017 Gaeta et al.
 2017/0066099 A1 3/2017 Nakamura
 2017/0114260 A1 4/2017 Bock et al.
 2017/0129075 A1 5/2017 Thurber et al.
 2017/0145274 A1 5/2017 Yener et al.
 2017/0158930 A1 6/2017 Iyengar
 2017/0225299 A1 8/2017 Keipert et al.
 2017/0247592 A1 8/2017 Bauer et al.
 2017/0335155 A1 11/2017 Czerepinski et al.
 2017/0335156 A1 11/2017 Bauer et al.
 2017/0342303 A1 11/2017 Stevenson et al.
 2017/0349797 A1 12/2017 Yener et al.
 2018/0002584 A1 1/2018 Yener et al.
 2018/0086957 A1 3/2018 Sahlin et al.
 2018/0155592 A1 6/2018 Josseaux et al.
 2018/0161960 A1 6/2018 Wilson et al.
 2018/0169837 A1 6/2018 Liu
 2018/0187057 A1 7/2018 Bujnowski et al.
 2018/0215975 A1 8/2018 Marazano et al.
 2018/0215976 A1 8/2018 Cotter et al.
 2018/0237675 A1 8/2018 Yener et al.
 2018/0318983 A1 11/2018 Wilson et al.
 2018/0327644 A1 11/2018 Bauer et al.
 2018/0370857 A1 12/2018 Marlin et al.
 2019/0022826 A1 1/2019 Franke et al.
 2019/0030684 A1 1/2019 Van et al.
 2019/0091835 A1 3/2019 Culler et al.
 2019/0119540 A1 4/2019 Colet et al.
 2019/0126436 A1 5/2019 Westberg et al.
 2019/0160630 A1 5/2019 Jiang et al.
 2019/0217442 A1 7/2019 Gaeta et al.
 2019/0217444 A1* 7/2019 Zhang B24D 11/02
 2019/0249052 A1 8/2019 Eckel et al.
 2019/0270182 A1 9/2019 Eckel et al.
 2019/0284461 A1 9/2019 Josseaux et al.
 2019/0309201 A1 10/2019 Dumont et al.
 2019/0322915 A1 10/2019 Jiwpanich et al.
 2019/0330505 A1 10/2019 Bujnowski et al.
 2019/0337124 A1 11/2019 Liu et al.
 2019/0338172 A1 11/2019 Erickson et al.
 2019/0338173 A1 11/2019 Yener et al.
 2019/0351531 A1 11/2019 Nelson et al.
 2019/0358776 A1 11/2019 Seth et al.
 2019/0366511 A1 12/2019 Huber
 2019/0382637 A1 12/2019 Braun et al.
 2020/0139512 A1 5/2020 Culler et al.
 2020/0148927 A1 5/2020 Arcona et al.
 2020/0156215 A1 5/2020 Jusuf et al.
 2020/0157396 A1 5/2020 Cotter et al.
 2020/0157397 A1 5/2020 Stevenson et al.
 2020/0199426 A1 6/2020 Yener et al.
 2020/0262031 A1 8/2020 Seth et al.
 2020/0308462 A1 10/2020 Bauer et al.
 2020/0391354 A1 12/2020 Marazano et al.
 2021/0024798 A1 1/2021 Czerepinski et al.
 2021/0087444 A1 3/2021 Stevenson et al.
 2021/0087445 A1 3/2021 Cotter et al.
 2021/0108117 A1 4/2021 Bauer et al.
 2021/0108118 A1 4/2021 Yener et al.
 2021/0130667 A1 5/2021 Arcona et al.
 2021/0197339 A1 7/2021 Marlin et al.
 2021/0198544 A1 7/2021 Marlin et al.
 2021/0198545 A1 7/2021 Marlin et al.
 2021/0332278 A1 10/2021 Iyengar
 2021/0395587 A1 12/2021 Yener et al.
 2022/0001512 A1 1/2022 Gaeta et al.
 2022/0025237 A1 1/2022 Sahlin et al.
 2023/0061952 A1 3/2023 Lentz et al.
 2023/0065541 A1 3/2023 Colet et al.
 2023/0096577 A1 3/2023 Cotter et al.

2023/0135441 A1 5/2023 Seth et al.
 2023/0193100 A1 6/2023 Josseaux et al.
 2023/0211466 A1 7/2023 Martone et al.
 2023/0211467 A1* 7/2023 Martone B24D 11/001
 451/539

2023/0211468 A1 7/2023 Martone et al.
 2023/0220255 A1 7/2023 Yuyang et al.
 2023/0220256 A1 7/2023 Bujnowski et al.
 2023/0265326 A1 8/2023 Adefris
 2023/0272254 A1 8/2023 Yener et al.
 2023/0294247 A1 9/2023 Liu et al.
 2023/0332030 A1 10/2023 Bauer et al.
 2023/0357617 A9 11/2023 Yener et al.
 2024/0116153 A1 4/2024 Martone et al.
 2024/0123574 A1 4/2024 Martone et al.
 2024/0141219 A1 5/2024 Bauer et al.
 2024/0198488 A1 6/2024 Marlin et al.

FOREIGN PATENT DOCUMENTS

CH 685051 A5 3/1995
 CN 1229007 A 7/2005
 CN 1774488 A 5/2006
 CN 101389466 A 3/2009
 CN 101970347 A 2/2011
 CN 101980836 A 2/2011
 CN 102281992 A 12/2011
 CN 103189164 A 7/2013
 CN 103842132 A 6/2014
 CN 102123837 B 7/2014
 CN 104125875 A 10/2014
 CN 104994995 A 10/2015
 CN 105622071 A 6/2016
 CN 105713568 A 6/2016
 DE 3923671 C2 2/1998
 DE 102012023688 A1 4/2014
 DE 202014101739 U1 6/2014
 DE 202014101741 U1 6/2014
 DE 102013202204 A1 8/2014
 DE 102013210158 A1 12/2014
 DE 102013210716 A1 12/2014
 DE 102013212598 A1 12/2014
 DE 102013212622 A1 12/2014
 DE 102013212634 A1 12/2014
 DE 102013212639 A1 12/2014
 DE 102013212644 A1 12/2014
 DE 102013212653 A1 12/2014
 DE 102013212654 A1 12/2014
 DE 102013212661 A1 12/2014
 DE 102013212666 A1 12/2014
 DE 102013212677 A1 12/2014
 DE 102013212680 A1 12/2014
 DE 102013212687 A1 12/2014
 DE 102013212690 A1 12/2014
 DE 102013212700 A1 12/2014
 DE 102014210836 A1 12/2014
 EP 0078896 A2 5/1983
 EP 0152768 A2 8/1985
 EP 0293163 A2 11/1988
 EP 0480133 A2 4/1992
 EP 0652919 A1 5/1995
 EP 0662110 A1 7/1995
 EP 0500369 B1 1/1996
 EP 0609864 B1 11/1996
 EP 0771769 5/1997
 EP 0812456 B1 12/1997
 EP 0651778 B1 5/1998
 EP 0614861 B1 5/2001
 EP 0931032 B1 7/2001
 EP 0833803 8/2001
 EP 1207015 A2 5/2002
 EP 1356152 A2 10/2003
 EP 1371451 A1 12/2003
 EP 1383631 A1 1/2004
 EP 1015181 B1 3/2004
 EP 1492845 A1 1/2005
 EP 1851007 11/2007
 EP 1960157 A1 8/2008
 EP 2176031 A1 4/2010

(56)

References Cited

FOREIGN PATENT DOCUMENTS

EP	2184134	A1	5/2010	JP	2017538588	A	12/2017
EP	2242618	A2	10/2010	JP	2018510073	A	4/2018
EP	2390056	A2	11/2011	KR	1019890014409	A	10/1989
EP	1800801	B1	3/2012	KR	1020020042840	A	6/2002
EP	2445982	A2	5/2012	KR	20140106713	A	9/2014
EP	2507016	A2	10/2012	NL	171464	B1	11/1982
EP	2537917	A1	12/2012	WO	94/02559	A1	2/1994
EP	2567784	A1	3/2013	WO	95/03370		2/1995
EP	2631286	A1	8/2013	WO	1995016756	A1	6/1995
EP	2692813	A1	2/2014	WO	1995017287	A1	6/1995
EP	2692814	A1	2/2014	WO	95/18192	A1	7/1995
EP	2692815	A1	2/2014	WO	95/20469	A1	8/1995
EP	2692816	A1	2/2014	WO	1996012776	A1	5/1996
EP	2692817	A1	2/2014	WO	1996014964	A1	5/1996
EP	2692818	A1	2/2014	WO	96/27189	A1	9/1996
EP	2692819	A1	2/2014	WO	97/11484	A1	3/1997
EP	2692820	A1	2/2014	WO	97/14536	A1	4/1997
EP	2692821	A1	2/2014	WO	99/06500	A1	2/1999
EP	2719752	A1	4/2014	WO	99/14016	A1	3/1999
EP	2012972	B1	6/2014	WO	99/38817	A1	8/1999
EP	2720676	B1	1/2018	WO	1999038817	A1	8/1999
EP	3319758	A1	5/2018	WO	99/54424	A1	10/1999
EP	3342839	A1	7/2018	WO	0064630	A1	11/2000
EP	3444313	B1	7/2020	WO	0114494	A1	3/2001
EP	3830211	A1	6/2021	WO	0123323	A1	4/2001
FR	2354373	A1	1/1978	WO	02097150	A2	12/2002
GB	986847	A	3/1965	WO	03087236	A1	10/2003
GB	1456765	A	11/1976	WO	2005080624	A1	9/2005
GB	1466054		3/1977	WO	2005112601	A2	12/2005
JP	53064890	A	6/1978	WO	2006027593	A3	3/2006
JP	60-006356	U	1/1985	WO	2006062597	A1	6/2006
JP	62002946	B	1/1987	WO	2007041538	A1	4/2007
JP	63036905	B	7/1988	WO	2009085578	A2	7/2009
JP	03079277	A	4/1991	WO	2009085841	A2	7/2009
JP	03-287687		12/1991	WO	2009098017	A1	8/2009
JP	05285833	A	11/1993	WO	2010077509	A1	7/2010
JP	06114739	A	4/1994	WO	2010085587	A1	7/2010
JP	07008474	B2	2/1995	WO	2010118440	A2	10/2010
JP	3030861	U	8/1996	WO	2010151201	A1	12/2010
JP	10113875	A	5/1998	WO	2011005425	A2	1/2011
JP	2779252	B2	7/1998	WO	2011019188	A1	2/2011
JP	10330734	A	12/1998	WO	2011068714	A2	6/2011
JP	H10315142	A	12/1998	WO	2011068724	A2	6/2011
JP	2957492	B2	10/1999	WO	2011087649	A2	7/2011
JP	2000091280	A	3/2000	WO	2011109188	A2	9/2011
JP	2000-336344	A	12/2000	WO	2011133438	A1	10/2011
JP	2000354967	A	12/2000	WO	2011139562	A2	11/2011
JP	3160084	B2	4/2001	WO	2011149625	A2	12/2011
JP	2001162541	A	6/2001	WO	2012018903	A2	2/2012
JP	3194269	B2	7/2001	WO	2012061016	A1	5/2012
JP	2001180930	A	7/2001	WO	2012061033	A2	5/2012
JP	2001207160	A	7/2001	WO	2012092590	A2	7/2012
JP	2001516652	A	10/2001	WO	2012092605	A3	7/2012
JP	2002-038131	A	2/2002	WO	2010070294	A1	8/2012
JP	2002210659	A	7/2002	WO	2012112305	A2	8/2012
JP	2003-049158	A	2/2003	WO	2012112322	A2	8/2012
JP	2004-510873	A	4/2004	WO	2012092590	A3	10/2012
JP	2004209624	A	7/2004	WO	2012140617	A1	10/2012
JP	2006130586	A	5/2006	WO	2012141905	A2	10/2012
JP	2006130636	A	5/2006	WO	2013003830	A2	1/2013
JP	2006159402	A	6/2006	WO	2013003831	A2	1/2013
JP	2006-192540	A	7/2006	WO	2013009484	A2	1/2013
JP	2006224201	A	8/2006	WO	2013036402	A1	3/2013
JP	2007-537891	A	12/2007	WO	2013040423	A2	3/2013
JP	2008132560	A	6/2008	WO	2013045251	A1	4/2013
JP	2008194761	A	8/2008	WO	2013049239	A1	4/2013
JP	2008531305	A	8/2008	WO	2013070576	A2	5/2013
JP	2012512046	A	5/2012	WO	2013101575	A1	7/2013
JP	2012512047	A	5/2012	WO	2013102170	A1	7/2013
JP	2012512048	A	5/2012	WO	2013102176	A1	7/2013
JP	2012530615	A	12/2012	WO	2013102177	A1	7/2013
JP	5238725	B2	7/2013	WO	2013106597	A1	7/2013
JP	5238726	B2	7/2013	WO	2013106602	A1	7/2013
JP	2014503367	A	2/2014	WO	2013149209	A1	10/2013
JP	2017518889	A	7/2017	WO	2013151745	A1	10/2013
				WO	2013177446	A1	11/2013
				WO	2013186146	A1	12/2013
				WO	2013188038	A1	12/2013
				WO	2014005120	A1	1/2014

(56)	References Cited			WO	2018057465	A1	3/2018
	FOREIGN PATENT DOCUMENTS			WO	2018057558	A1	3/2018
				WO	2018063902	A1	4/2018
				WO	2018063958	A1	4/2018
				WO	2018063960	A1	4/2018
WO	2014020068	A1	2/2014	WO	2018063962	A1	4/2018
WO	2014020075	A1	2/2014	WO	2018064642	A1	4/2018
WO	2014022453	A1	2/2014	WO	2018080703	A1	5/2018
WO	2014022462	A1	2/2014	WO	2018080704	A1	5/2018
WO	2014022465	A1	2/2014	WO	2018080705	A1	5/2018
WO	2014161001	A1	2/2014	WO	2018080755	A1	5/2018
WO	2014057273	A1	4/2014	WO	2018080756	A1	5/2018
WO	2014062701	A1	4/2014	WO	2018080765	A1	5/2018
WO	2014070468	A1	5/2014	WO	2018080778	A1	5/2018
WO	2014106173	A1	7/2014	WO	2018080784	A1	5/2018
WO	2014106211	A1	7/2014	WO	2018081246	A1	5/2018
WO	2014124554	A1	8/2014	WO	2018118688	A1	6/2018
WO	2014137972	A1	9/2014	WO	2018118690	A1	6/2018
WO	2014140689	A1	9/2014	WO	2018118695	A1	6/2018
WO	2014165390	A1	10/2014	WO	2018118699	A1	6/2018
WO	2014176108	A1	10/2014	WO	2018134732	A1	7/2018
WO	2014206739	A1	12/2014	WO	2018136268	A1	7/2018
WO	2014206890	A1	12/2014	WO	2018136269	A1	7/2018
WO	2014206967	A1	12/2014	WO	2018136271	A1	7/2018
WO	2014209567	A1	12/2014	WO	2018172193	A1	9/2018
WO	2014210160	A1	12/2014	WO	2018207145	A1	11/2018
WO	2014210442	A1	12/2014	WO	2018226912	A1	12/2018
WO	2014210532	A1	12/2014	WO	2018236989	A1	12/2018
WO	2014210568	A1	12/2014	WO	2019001908	A1	1/2019
WO	2015050781	A1	4/2015	WO	2019069157	A1	4/2019
WO	2015073346	A1	5/2015	WO	2019102312	A1	5/2019
WO	2015048768	A9	6/2015	WO	2019102328	A1	5/2019
WO	2015088953	A1	6/2015	WO	2019102329	A1	5/2019
WO	2015089527	A1	6/2015	WO	2019102330	A1	5/2019
WO	2015089528	A1	6/2015	WO	2019102331	A1	5/2019
WO	2015089529	A1	6/2015	WO	2019108805	A2	6/2019
WO	2015100018	A1	7/2015	WO	2021161129	A1	8/2019
WO	2015100020	A1	7/2015	WO	2019167022	A1	9/2019
WO	2015100220	A1	7/2015	WO	2019197948	A1	10/2019
WO	2015102992	A1	7/2015	WO	2019207415	A1	10/2019
WO	2015112379	A1	7/2015	WO	2019207416	A1	10/2019
WO	2015130487	A1	9/2015	WO	2019207417	A1	10/2019
WO	2015143461	A1	10/2015	WO	2019207423	A1	10/2019
WO	2015158009	A1	10/2015	WO	2019215571	A1	11/2019
WO	2015160854	A1	10/2015	WO	2020025270	A1	2/2020
WO	2015160855	A1	10/2015	WO	2020035764	A1	2/2020
WO	2015160857	A1	10/2015	WO	2020075005	A1	4/2020
WO	2015164211	A1	10/2015	WO	2020079522	A1	4/2020
WO	2015165122	A1	11/2015	WO	2020084382	A1	4/2020
WO	2015167910	A1	11/2015	WO	2020084483	A1	4/2020
WO	2015179335	A1	11/2015	WO	2020089741	A1	5/2020
WO	2015180005	A1	12/2015	WO	2020115685	A1	6/2020
WO	2015184355	A1	12/2015	WO	2020128708	A1	6/2020
WO	2016028683	A1	2/2016	WO	2020128716	A1	6/2020
WO	2016044158	A1	3/2016	WO	2020128717	A1	6/2020
WO	2016064726	A1	4/2016	WO	2020128719	A1	6/2020
WO	2016089675	A1	6/2016	WO	2020128720	A2	6/2020
WO	2016105469	A1	6/2016	WO	2020128752	A1	6/2020
WO	2016105474	A1	6/2016	WO	2020128779	A2	6/2020
WO	2016160357	A1	10/2016	WO	2020128780	A1	6/2020
WO	2016161157	A1	10/2016	WO	2020128781	A1	6/2020
WO	2016161170	A1	10/2016	WO	2020128783	A1	6/2020
WO	2016167967	A1	10/2016	WO	2020128787	A1	6/2020
WO	2016187570	A1	11/2016	WO	2020128794	A1	6/2020
WO	2016196795	A1	12/2016	WO	2020128833	A1	6/2020
WO	2016201104	A1	12/2016	WO	2020128838	A1	6/2020
WO	2016205133	A1	12/2016	WO	2020128842	A1	6/2020
WO	2016205267	A1	12/2016	WO	2020128844	A1	6/2020
WO	2016210057	A1	12/2016	WO	2020128845	A1	6/2020
WO	2017007703	A1	1/2017	WO	2020128852	A1	6/2020
WO	2017007714	A1	1/2017	WO	2020128853	A1	6/2020
WO	2017062482	A1	4/2017	WO	2020128856	A1	6/2020
WO	2017083249	A1	5/2017	WO	2020212788	A1	10/2020
WO	2017083255	A1	5/2017	WO	2021009600	A1	1/2021
WO	2016105543	A9	9/2017	WO	2021014271	A1	1/2021
WO	2017151498	A1	9/2017	WO	2021074756	A1	4/2021
WO	2017197002	A1	11/2017	WO	2021074768	A1	4/2021
WO	2017197006	A1	11/2017	WO	2021079331	A1	4/2021
WO	2018010730	A2	1/2018	WO	2021081571	A1	5/2021
WO	2018026669	A1	2/2018				

(56)

References Cited

FOREIGN PATENT DOCUMENTS

WO	2021105030	A1	6/2021
WO	2021116883	A1	6/2021
WO	2021133876	A1	7/2021
WO	2021133888	A1	7/2021
WO	2021133901	A1	7/2021
WO	2021137092	A1	7/2021
WO	2021179025	A1	9/2021
WO	2021186326	A1	9/2021
WO	2021214576	A1	10/2021
WO	2021214605	A1	10/2021
WO	2021234540	A1	11/2021
WO	2022022905	A1	2/2022
WO	2022022906	A1	2/2022
WO	2022229744	A1	11/2022
WO	2023130051	A1	7/2023
WO	2023130052	A1	7/2023
WO	2023130053	A1	7/2023
WO	2023209518	A1	11/2023
WO	2024127255	A1	6/2024

OTHER PUBLICATIONS

Austin, Benson M., "Thick-Film Screen Printing," Solid State Technology, Jun. 1969, pp. 53-58.

Avril, Nicolas Joseph "Manufacturing Glass-Fiber Reinforcement for Grinding Wheels" Massachusetts Institute of Technology, Feb. 1996, 105 pages.

Bacher, Rudolph J., "High Resolution Thick Film Printing," E.I. du Pont de Nemours & Company, Inc., Proceedings of the International Symposium on Microelectronics, 1986, pp. 576-581.

Besse, John R., "Understanding and controlling wheel truing and dressing forces when rotary plunge dressing," Cutting Tool Engineering, Jun. 2012, vol. 64, Issue 6, 4 pages.

Ciccotti, M. et al., "Complex dynamics in the peeling of an adhesive tape," International Journal of Adhesion & Adhesives 24 (2004) pp. 143-151.

Dupont, "Kevlar Aramid Pulp", Copyright 2011, DuPont, Accessed: Sep. 18, 2013, 2 pages.

Wu, J. et al., "Friction and Wear Properties of Kevlar Pulp Reinforced Epoxy Composite under Dry Sliding Condition", Tribology Letters, vol. 22, No. 3, Jun. 2006, pp. 259-263, Abstract only.

J. European Ceramic Society 31 (2011) 2073-2081, Abstract only.

Riemer, Dietrich E., "Analytical Engineering Model of the Screen Printing Process: Part II," Solid State Technology, Sep. 1988, pp. 85-90.

Miller, L.F., "Paste Transfer in the Screening Process," Solid State Technology, Jun. 1969, pp. 46-52.

Morgan, P. et al., "Ceramic Composites of Monazite and Alumina," J. Am. Ceram. Soc., 78, 1995, 1553-63.

Riemer, Dietrich E., "Analytical Engineering Model of the Screen Printing Process: Part I," Solid State Technology, Aug. 1988, pp. 107-111.

Winter Catalogue No. 5, Dressing tools, Winter diamond tools for dressing grinding wheels, Edition Year: 2010, 140 pages.

Badger, Jeffrey, "Evaluation of Triangular, Engineered-Shape Ceramic Abrasive in Cutting Discs," Supplement to the Welding Journal, Apr. 2014, vol. 93, pp. 107-s to 115-s.

3M Cubitron II Abrasive Belts Brochure, Shaping the Future, Jan. 2011, 6 pages.

Vanstrum et al., Precisely Shaped Grain (PSG): 3M's Innovation in Abrasive Grain Technology, date unknown, 1 page.

Graf, "Cubitron II: Precision-Shaped Grain (PSG) Turns the Concept of Gear Grinding Upside Down," gearsolutions.com, May 2014, pp. 36-44.

Dow Machine Tool Accessories, Grinding & Surface Finishing, www.lmta.com, Nov. 2014, 72 pages.

VSM Actirox Fibre Discs, The Latest Generation of Abrasives for Maximum Stock Removal [PDF] VSM Abrasives Ltd., Apr. 2019 [retrieved on May 15, 2019], 8 pages. Retrieved from https://uk.vsmabrasives.com/fileadmin/user_upload/ACTIROX/VSM-ACTIROX-EN.pdf.

Kumar et al., "Composites by rapid prototyping technology", Material & Design, Feb. 2010, vol. 31, Issue 2, pp. 850-856.

Lewis et al., "Direct Ink Writing of Three-Dimensional Ceramic Structures", Journal of the American Ceramic Society, US, Nov. 30, 2006, vol. 89, Issue 12, pp. 3599-3609.

International Search Report with regard to International application No. PCT/US2017/031998, dated Aug. 21, 2017.

International Search Report with regard to International application No. PCT/US2017/031992, dated Aug. 21, 2017.

International Search Report and Written Opinion for Application No. PCT/US2016/036701, dated Sep. 1, 2016, 12 pages.

Brewer, L. et al., Journal of Materials Research, 1999, vol. 14, No. 10, pp. 3907-3912, Abstract only.

Lewis et al., "Direct Ink Writing of 3D Functional Materials", Advanced Functional Materials, 2006, 16, pp. 2193-2204.

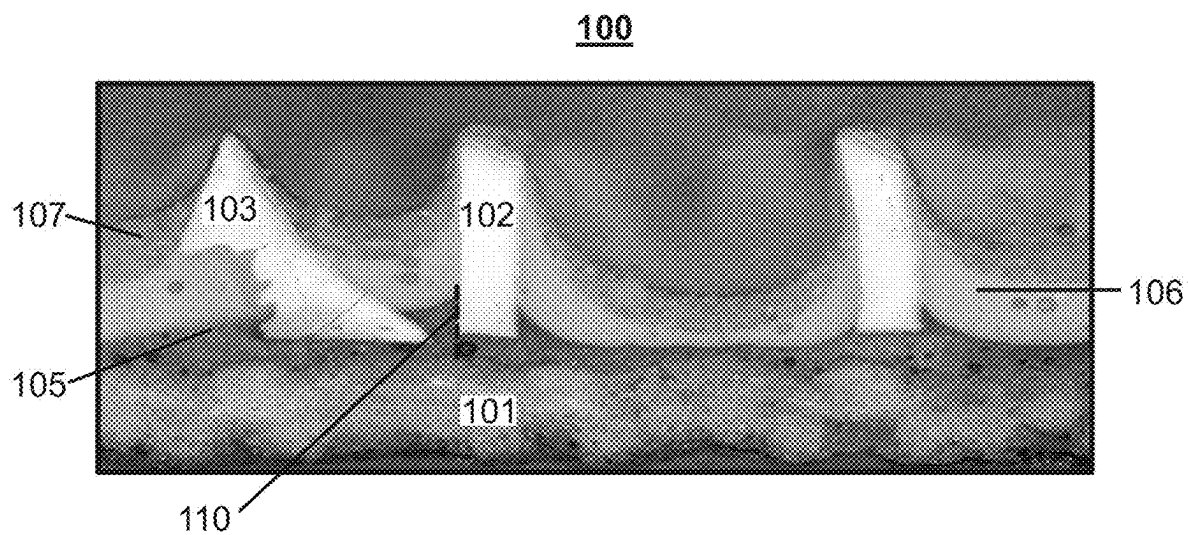
International Search Report and Written Opinion for Application No. PCT/US2020/066817, mailed Apr. 15, 2021, 11 pages.

International Search Report and Written Opinion for Application No. PCT/US2014/058378, mailed Jan. 29, 2015, 18 pages.

International Search Report and Written Opinion for Application No. PCT/US2015/025825) mailed Jul. 16, 2015, 13 pages.

International Search Report and Written Opinion for PCT/US2022/082599 dated Apr. 25, 2023, 12 pages.

* cited by examiner

**FIG. 1**

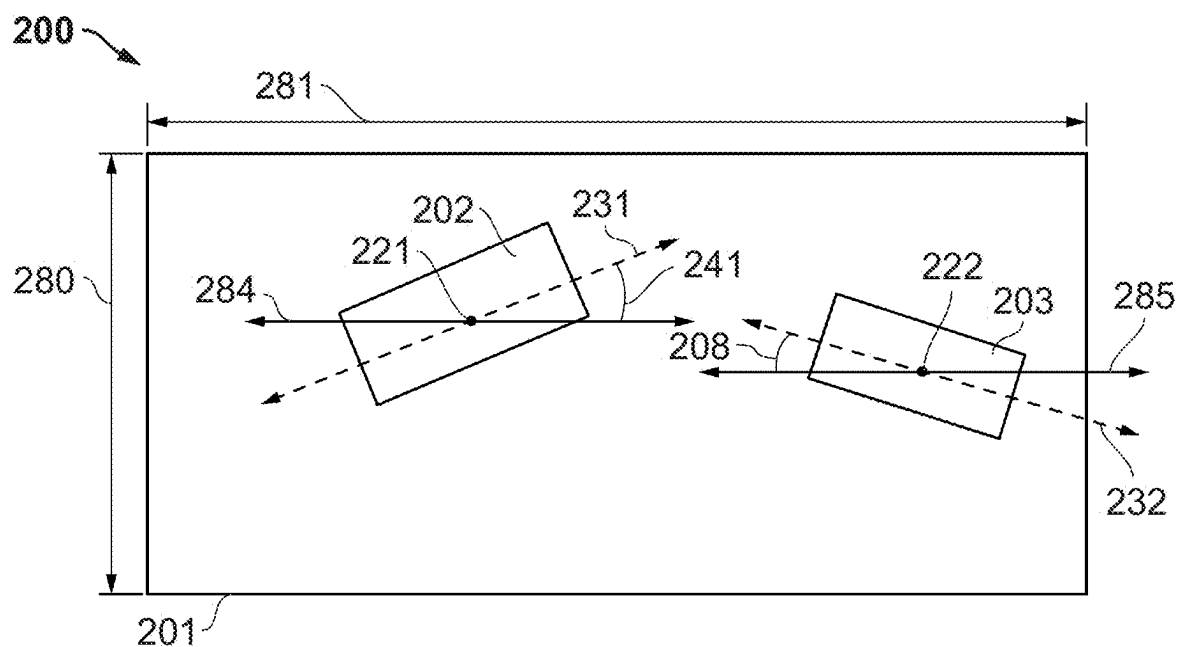


FIG. 2A

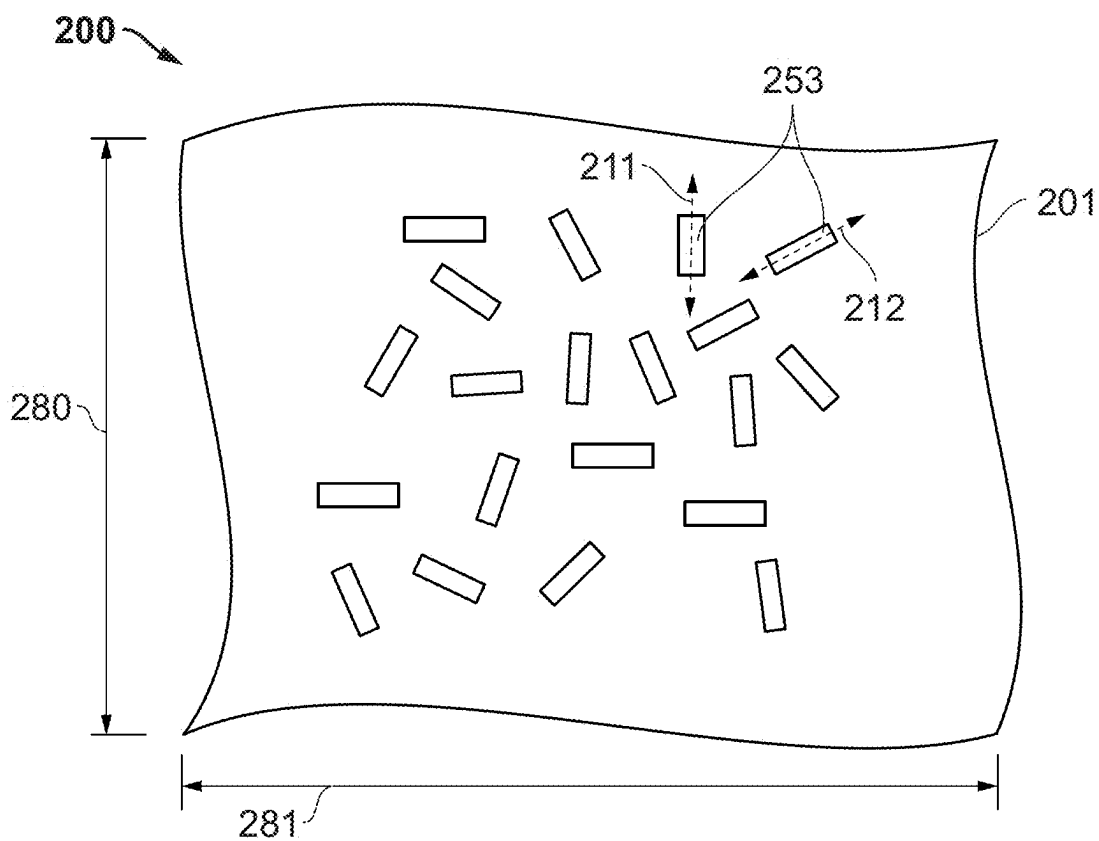


FIG. 2B

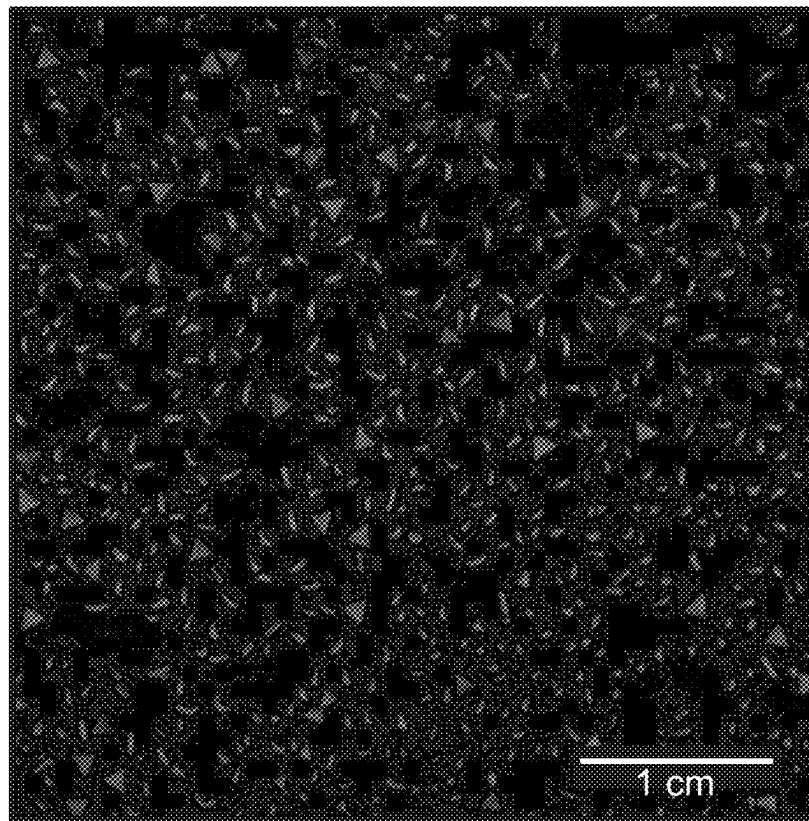


FIG. 3

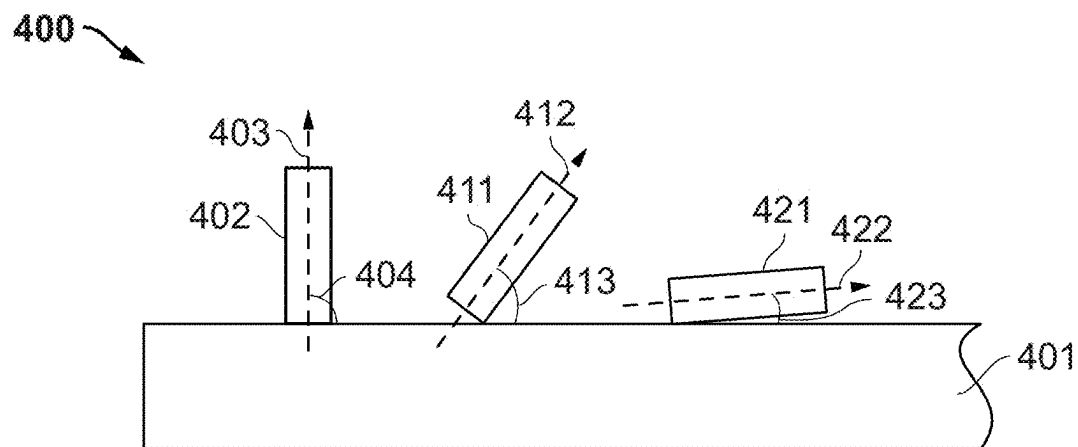


FIG. 4A

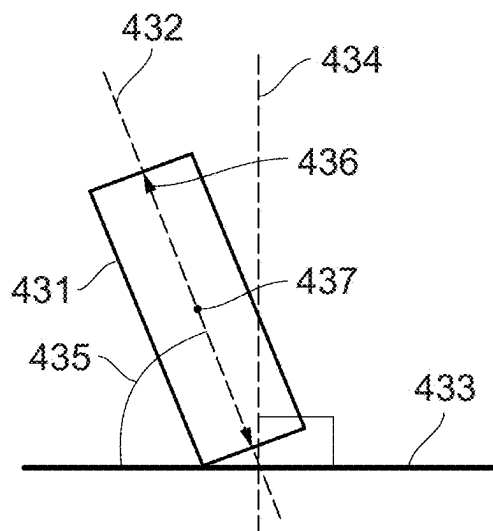


FIG. 4B

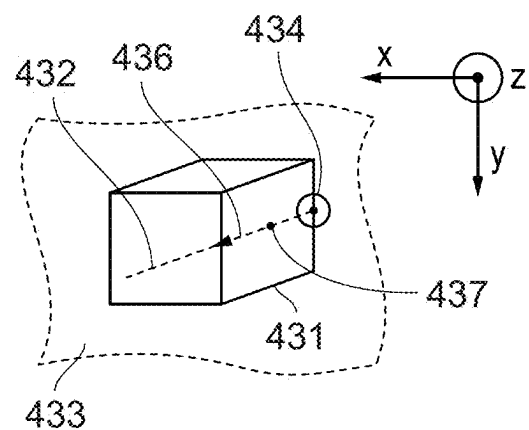


FIG. 4C

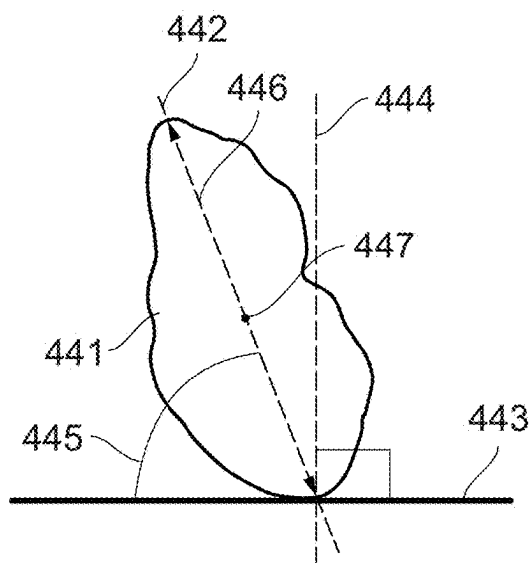


FIG. 4D

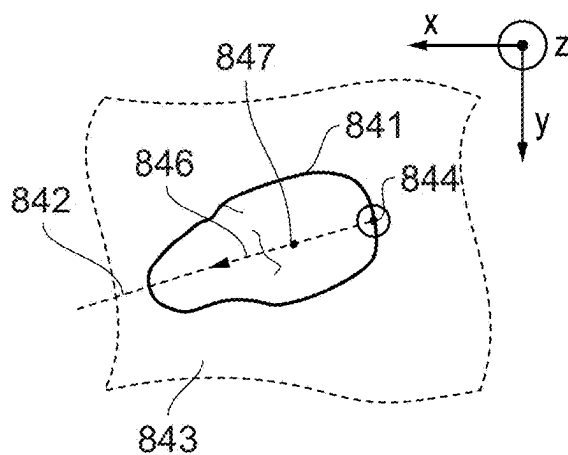


FIG. 4E

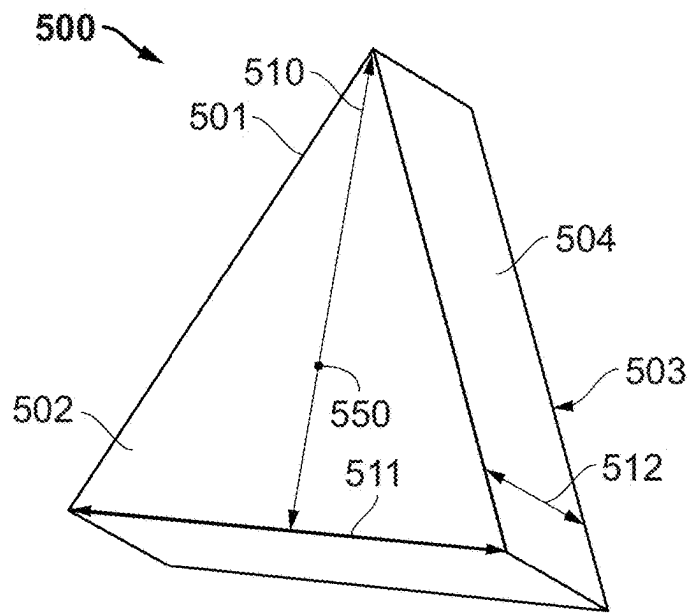


FIG. 5A

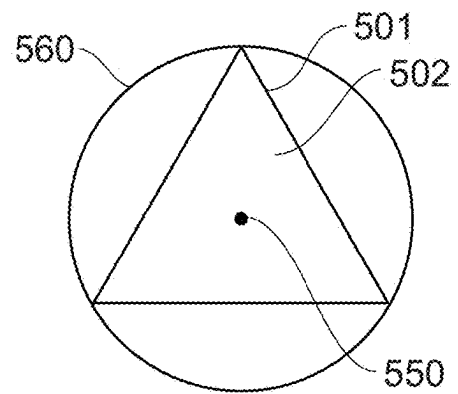


FIG. 5B

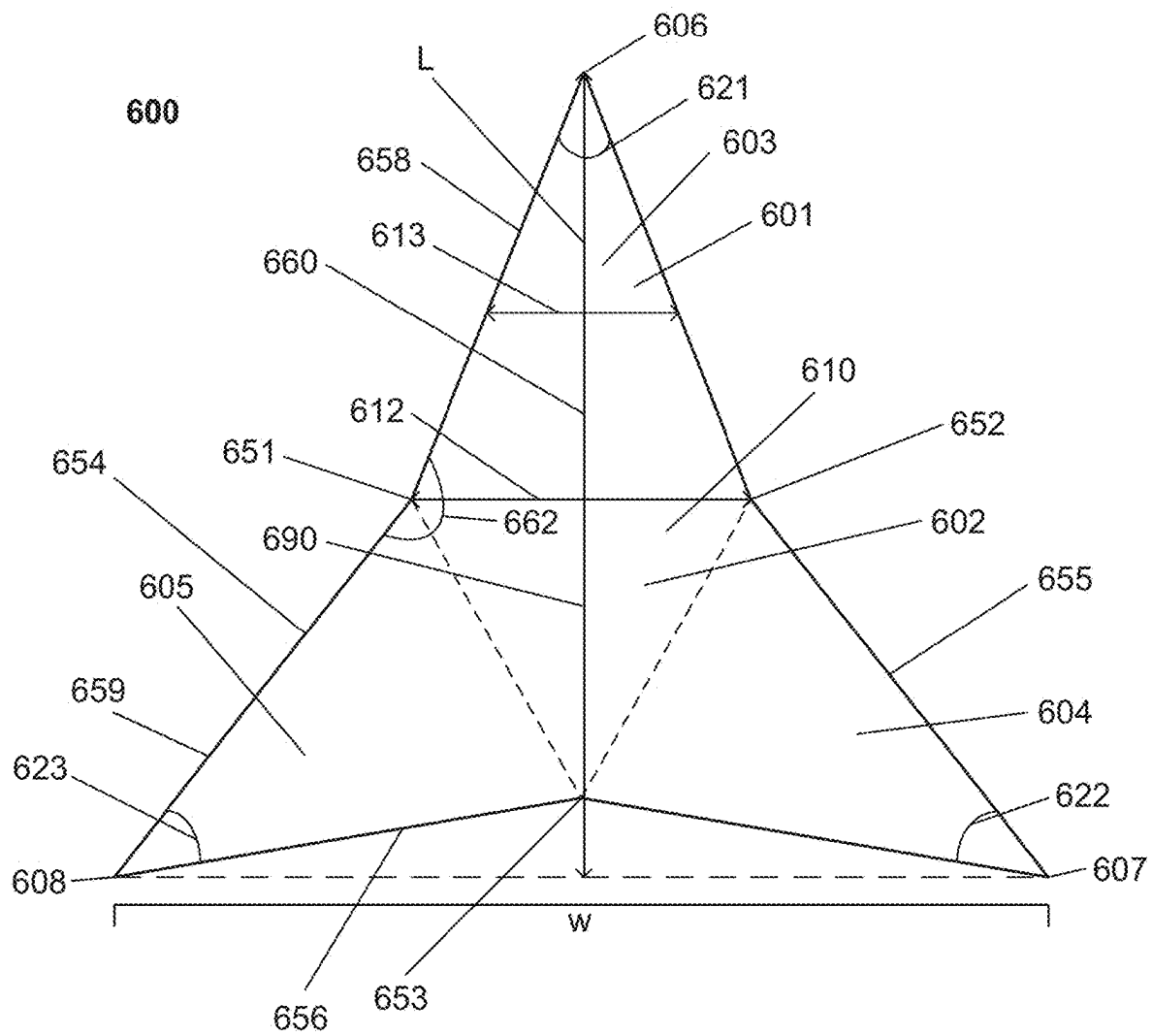
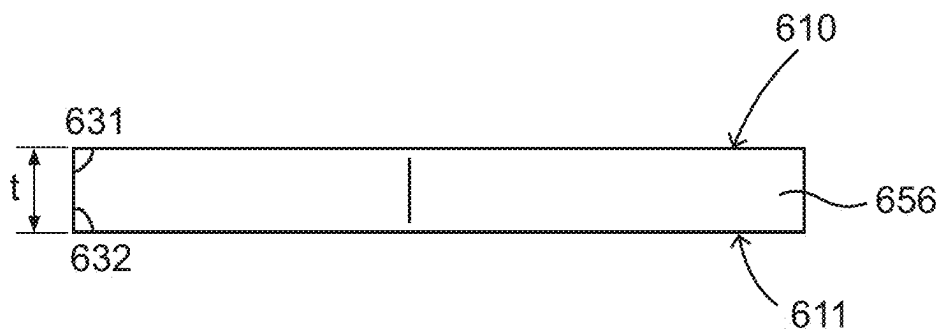


FIG. 6A

**FIG. 6B**

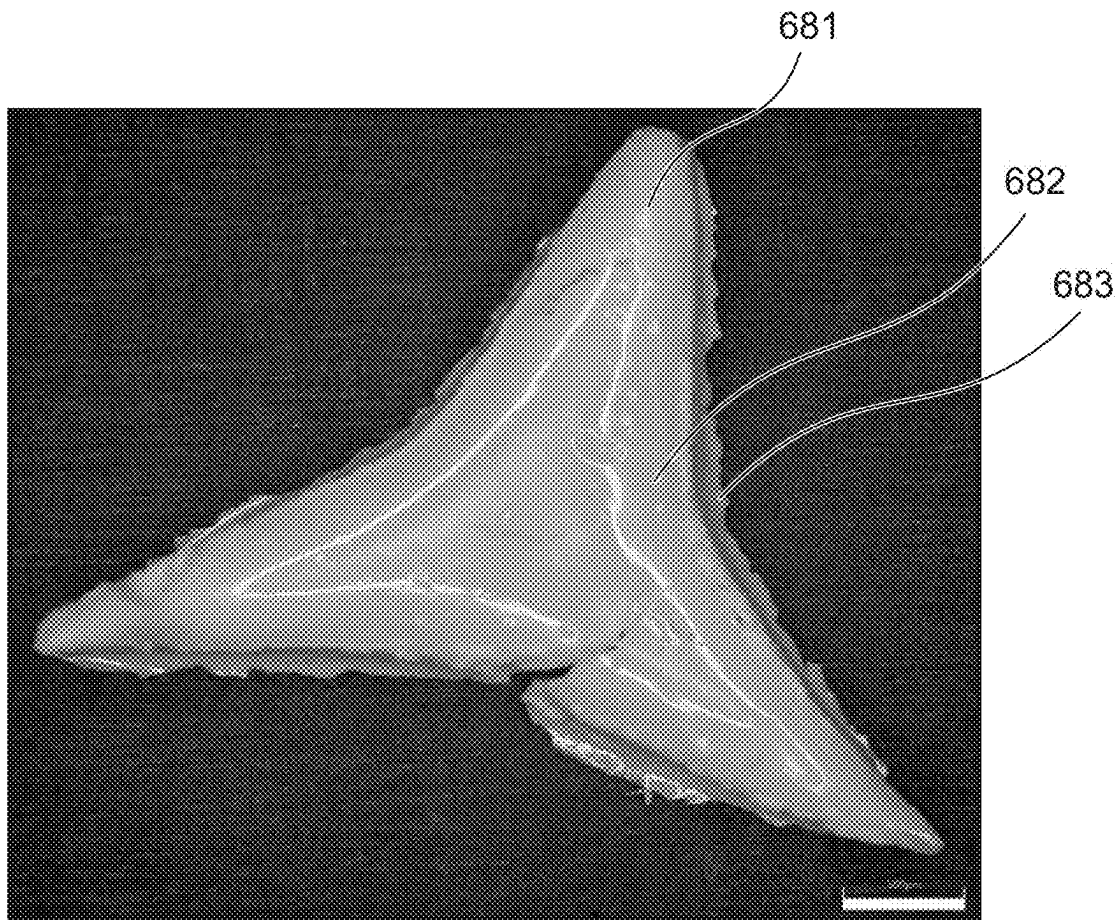
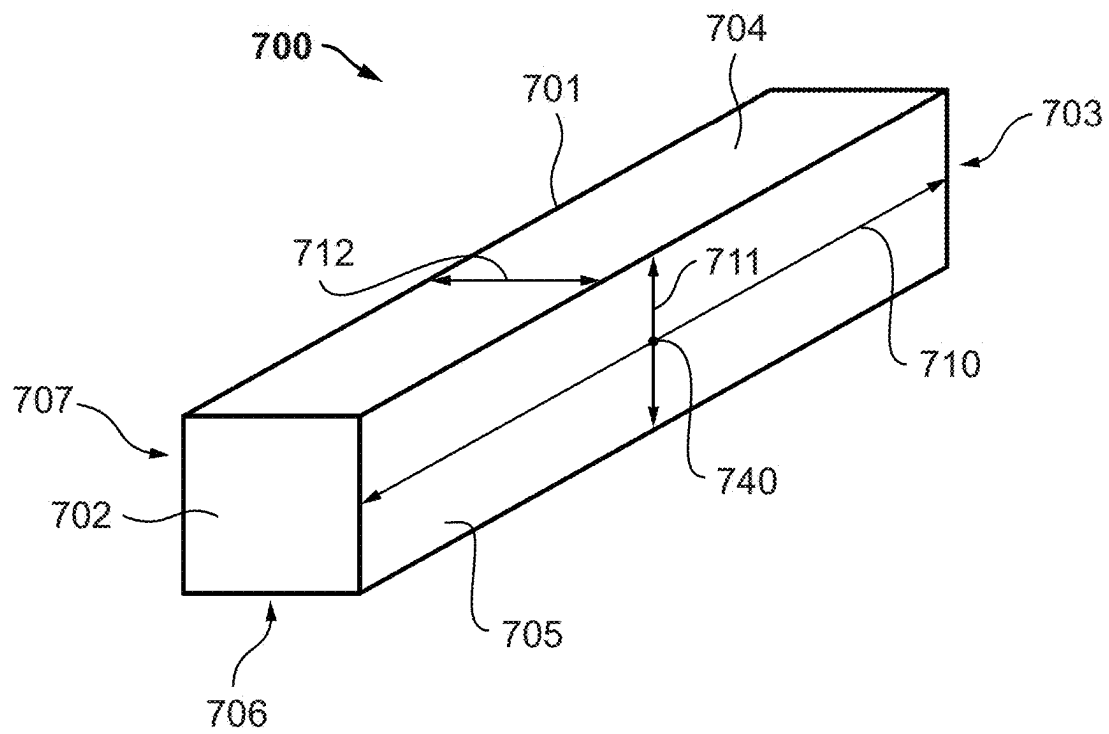


FIG. 6C

**FIG. 7**

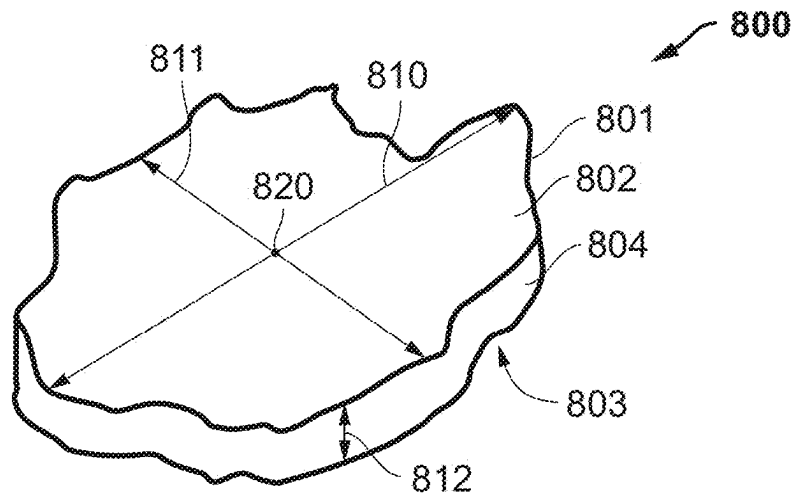


FIG. 8A

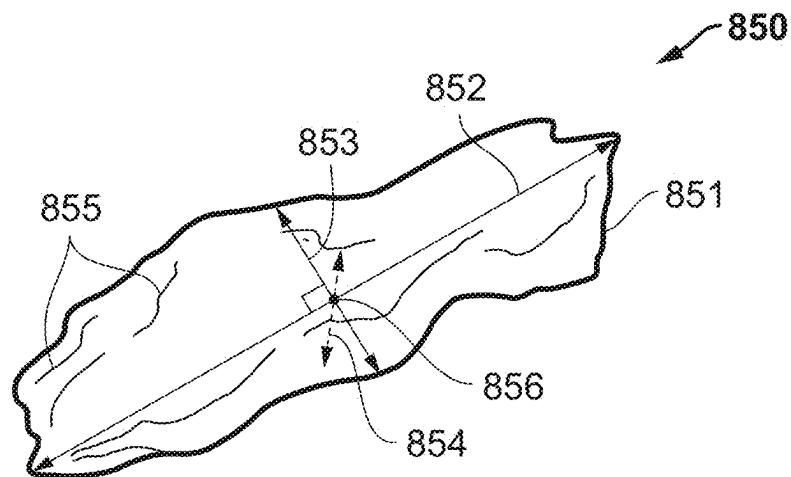
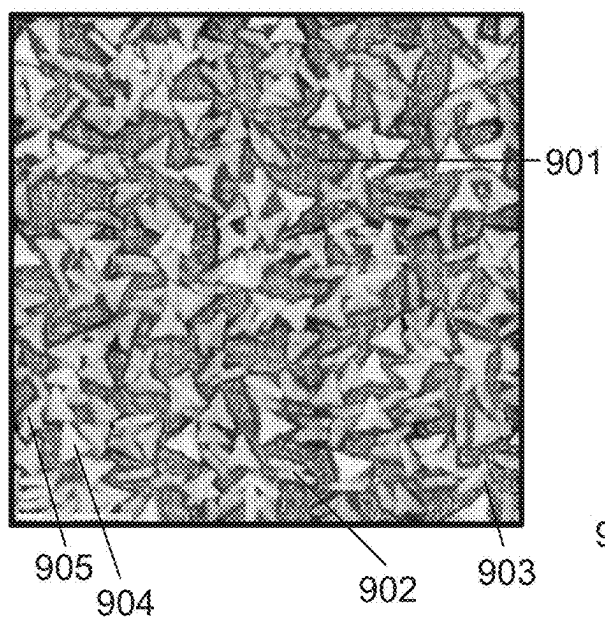
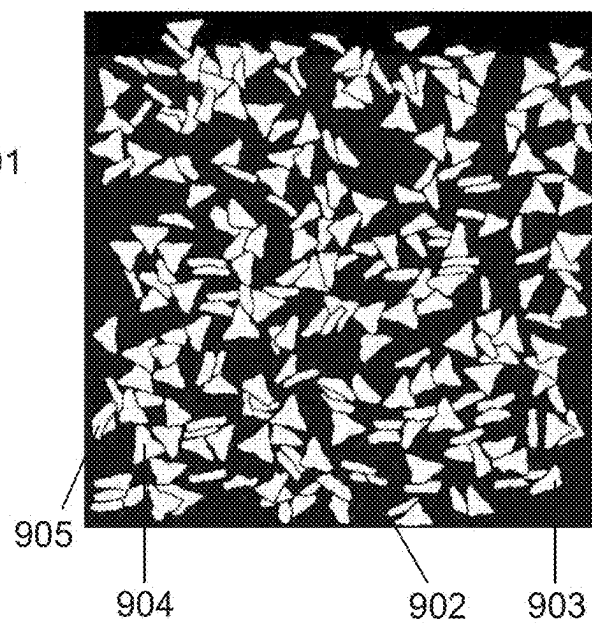
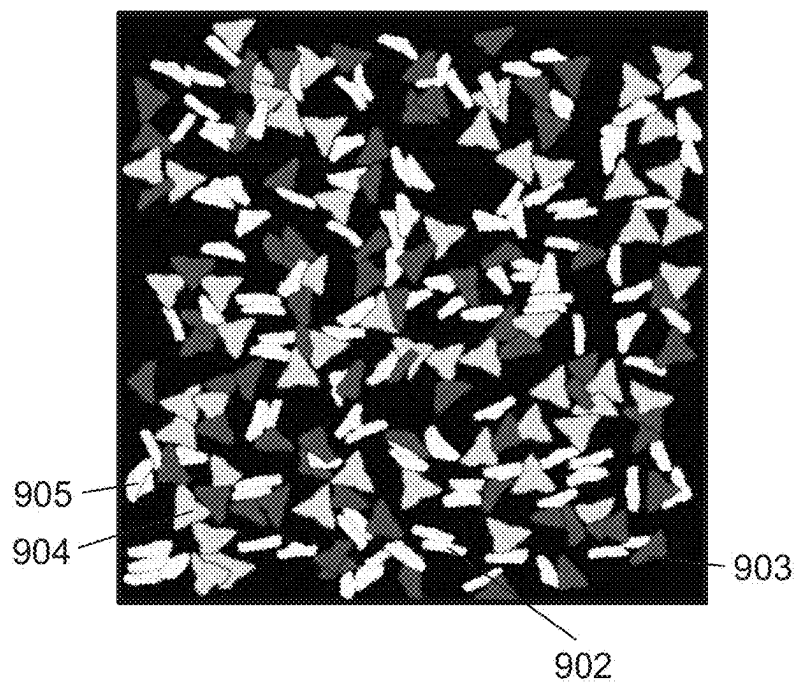


FIG. 8B

**FIG. 9A****FIG. 9B****FIG. 9C**

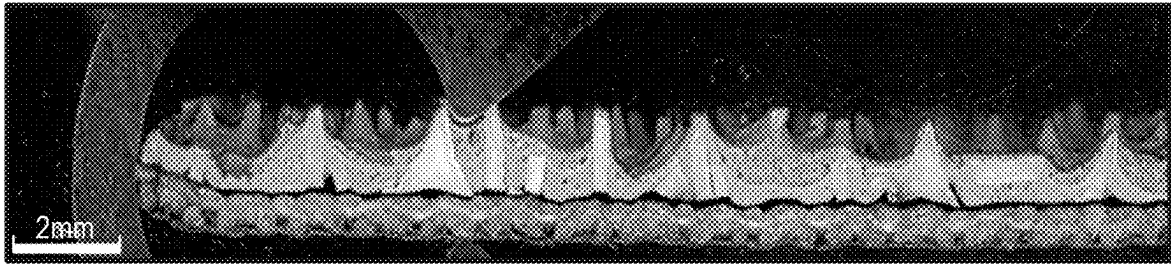


FIG. 10

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ABRASIVE ARTICLES AND METHODS OF FORMING SAME

CROSS-REFERENCE TO RELATED APPLICATION(S)

This application is a continuation application of and claims priority under 35 U.S.C. § 120 to U.S. patent application Ser. No. 18/148,403, entitled “ABRASIVE ARTICLES AND METHODS OF FORMING SAME,” by Anthony MARTONE et al., filed Dec. 29, 2022, which claims priority under 35 U.S.C. § 119(e) to U.S. Provisional Patent Application No. 63/266,269, entitled “ABRASIVE ARTICLES AND METHODS OF FORMING SAME,” by Anthony MARTONE et al., filed Dec. 30, 2021, both of which are assigned to the current assignees hereof and incorporated herein by reference in their entirety.

BACKGROUND

Field of the Disclosure

The following is directed to abrasive articles, and, in particular, coated abrasive articles and methods of forming coated abrasive articles.

SUMMARY

According to one aspect, an abrasive article includes a backing; a make coat overlying the backing; and a plurality of abrasive particles overlying the backing and at least partially contained in the make coat; and a make coat thickness ratio (T_g/T_a) of not greater than 1.5, wherein T_g is the average thickness of the make coat at the sides of the abrasive particles and the T_a is the average thickness of the make coat.

BRIEF DESCRIPTION OF THE DRAWINGS

The present disclosure may be better understood, and its numerous features and advantages made apparent to those skilled in the art by referencing the accompanying drawings.

FIG. 1 includes a cross sectional image of a section of an abrasive article according to an embodiment.

FIG. 2A includes a top-view illustration of a portion of a coated abrasive article according to an embodiment.

FIG. 2B includes a top-view illustration of a portion of a coated abrasive article according to an embodiment.

FIG. 3 includes an exemplary image of a portion of an abrasive article having abrasive particles in a random orientation.

FIG. 4A includes a side-view illustration of abrasive particles on a backing according to an embodiment.

FIG. 4B includes a side-view illustration of a particle on a backing having a tilt angle according to an embodiment.

FIG. 4C includes a top-down illustration of the particle of FIG. 4B.

FIG. 4D includes a side-view illustration of a particle on a backing having a tilt angle according to an embodiment.

FIG. 4E includes a top-down illustration of the particle of FIG. 4D.

FIG. 5A includes a perspective view illustration of a shaped abrasive particle according to an embodiment.

FIG. 5B includes a top-down view illustration of a shaped abrasive particle according to an embodiment.

FIG. 6A includes an image of a 3-PT star-shaped abrasive particle.

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FIG. 6B includes an illustration of a side view of the shaped abrasive particle of FIG. 6A.

FIG. 6C includes an image of 3-PT star-shaped abrasive particle.

FIG. 7 includes a perspective view illustration of a shaped abrasive particle according to an embodiment.

FIG. 8A includes a perspective view illustration of a controlled height abrasive particle according to an embodiment.

FIG. 8B includes a perspective view illustration of a non-shaped particle according to an embodiment.

FIG. 9A includes a top-down view of a coated abrasive article.

FIG. 9B includes the image of FIG. 9A edited to highlight the abrasive grains.

FIG. 9C includes the image of FIG. 9B further edited to color code the grains by orientation.

FIG. 10 includes an image of an abrasive article with a colored make layer.

DETAILED DESCRIPTION

The following is directed to methods of forming abrasive articles, such as fixed abrasive articles, and more particularly, coated abrasive articles. The abrasive articles may be used in a variety of material removal operations for a variety of work pieces.

FIG. 1 includes an image of a coated abrasive article **100** according to an embodiment. As shown in FIG. 1, the coated abrasive article **100** can include a backing **101**. The coated abrasive article **100** can also include an adhesive layer such as make coat **105** overlying the backing. The coated abrasive article **100** can further include a plurality of abrasive particles **102** and **103**. The coated abrasive article can also include a size coat **106** and a super size coat **107**. The make coat **105** can have an average thickness, T_a . The make coat **105** can also include an average thickness at the sides of the abrasive particles, T_g . An example make coat thickness at the side of the abrasive particles can be seen as dotted line **110** in FIG. 1.

In an embodiment, the make coat can comprise a particular material that may facilitate improved manufacturing or performance of the abrasive article. In an embodiment, the make coat can include Wollastonite, PF resin, water, or a combination thereof.

In an embodiment, the make coat can have a particular viscosity that may facilitate improved manufacturing or performance of the abrasive article. In an embodiment, the viscosity can be at least 3500 cps or at least 3750 cps or at least 4000 cps or at least 4250 cps or at least 4500 cps or at least 4750 cps or at least 5000 cps or at least 5250 cps. In an embodiment, the viscosity can be no greater than 7000 cps or no greater than 6750 cps or no greater than 6500 cps or no greater than 6250 cps or no greater than 6000 cps or no greater than 5750. It will be appreciated that the make coat viscosity can be between any of the above mentioned minimum and maximum values noted above, including, for example, but not limited to, at least 3500 cps and not greater than 7000 cps or at least 5000 cps and not greater than 6000 cps.

Average make coat thickness can be measured according to the following procedure. Abrasive articles are cut through the middle to reveal a cross section. The articles are then cut into 2-inch segments and mounted on an epoxy puck. Two 2-inch segments are then imaged, and the make layer is identified by coloring in the layer using the imaging software. FIG. 10 includes an example image of an abrasive

article include a colored make layer. Image analysis is used to overlay vertical gridlines, and the line segments overlapping the make layer are identified and isolated. Each line segment corresponds to a make coat thickness measurement. The average of all segments is taken. Approximately 150-200 overlapping line segments were made per two-inch sample segment, resulting in over 300 measurements for each sample.

Average make coat thickness near standing grains can be measured according to the following procedure. The same cross-sectional images for average make coat thickness can also be used for average make coat thickness near standing grains. Only standing grains showing their cross-sectional rectangular area with their short side in contact with the make coat are considered. For example, in FIG. 1, grain **102** would be considered but grain **103** would not. Additionally, only isolated grains were considered. Standing grains in contact with another grain were not considered for average make coat thickness near standing grains measurements. Measurements were made from the highest point of make contacting the grain side down to the lowest point of make contacting the backing on both sides of grain. The line of measurement is made perpendicular to the backing plane.

In an embodiment, the coated abrasive article can have a make coat of a particular average thickness that may facilitate improved performance and/or manufacturing of the abrasive article. In an embodiment, the average thickness of the make coat, Ta, can be at least 50 microns or at least 60 microns or at least 70 microns or at least 80 microns or at least 90 microns or at least 100 microns or at least 110 microns or at least 120 microns or at least 130 microns or at least 140 microns or at least 150 microns or at least 160 microns. In another embodiment, the average thickness of the make coat, Ta, can be not greater than 1 mm or not greater than 800 microns or not greater than 700 microns or not greater than 600 microns or not greater than 500 microns or not greater than 400 microns or not greater than 300 microns or not greater than 275 microns or not greater than 250 microns or not greater than 225 microns or not greater than 200 microns. It will be appreciated that Ta can be between any of the minimum and maximum values noted above, including, for example, but not limited to, at least 50 microns, and no greater than 800 microns, or at least 80 microns and no greater than 300 microns.

In an embodiment, the coated abrasive article can have a make coat of a particular average thickness at the sides of the abrasive particles, Tg, that may facilitate improved performance and/or manufacturing of the abrasive article. In an embodiment, Tg can be at least 50 microns or at least 60 microns or at least 70 microns or at least 80 microns or at least 90 microns or at least 100 microns or at least 110 microns or at least 120 microns or at least 130 microns or at least 140 microns or at least 150 microns. In another embodiment, Tg can be not greater than 1 mm or not greater than 800 microns or not greater than 700 microns or not greater than 600 microns or not greater than 500 microns or not greater than 400 microns or not greater than 300 microns. It will be appreciated that Tg can be between any of the minimum and maximum values noted above, including, for example, but not limited to, at least 50 microns, and no greater than 800 microns, or at least 80 microns and no greater than 300 microns.

In an embodiment, the coated abrasive article can have a make coat of a particular thickness standard deviation at the sides of the abrasive particles, STDTg, that may facilitate improved performance and/or manufacturing of the abrasive article. In an embodiment, STDTg can be at least 1 micron

or at least 5 microns or at least 10 microns or at least 15 microns or at least 20 microns or at least 25 microns or at least 30 microns. In another embodiment, STDTg can be not greater than 100 microns or not greater than 90 microns or not greater than 85 microns or not greater than 80 microns or not greater than 75 microns or not greater than 70 microns or not greater than 65 microns or not greater than 60 microns or not greater than 55 microns or not greater than 50 microns or not greater than 45 microns or not greater than 40 microns or not greater than 35 microns or not greater than 30 microns. It will be appreciated that STDTg can be between any of the minimum and maximum values noted above, including, for example, but not limited to, at least 5 microns, and no greater than 100 microns, or at least 10 microns and no greater than 45 microns.

In an embodiment, the coated abrasive article can have a make coat of a particular thickness ratio, Tg/Ta, that may facilitate improved performance and/or manufacturing of the abrasive article. In an embodiment, Tg/Ta can be not greater than 1.45 or not greater than 1.43 or not greater than 1.40 or not greater than 1.38 or not greater than 1.35 or not greater than 1.33 or not greater than 1.30 or not greater than 1.28 or not greater than 1.25 or not greater than 1.23 or not greater than 1.20 or not greater than 1.18 or not greater than 1.15 or not greater than 1.13 or not greater than 1.10 or not greater than 1.08 or not greater than 1.05 or not greater than 1.03. In another embodiment, Tg/Ta can be at least 0.70 or at least 0.80 or at least 0.90 or at least 0.98 or at least 1.00 or at least 1.03 or at least 1.05 or at least 1.08. It will be appreciated that Tg/Ta can be between any of the minimum and maximum values noted above, including, for example, but not limited to, at least 0.8, and no greater than 1.45, or at least 0.98 and no greater than 1.20.

FIG. 2A includes an illustration of a portion of a coated abrasive article **200** according to an embodiment. As shown in FIG. 2, the coated abrasive article **200** can include a backing **201** having a longitudinal axis **280** and a lateral axis **281**. The abrasive article **200** can include a backing **201** having a major surface and an abrasive layer forming an abrasive surface overlying the major surface of the backing. The abrasive layer can form a single layer of abrasive particles **202** and **203** adhered to the major surface of the backing. FIG. 2B also includes an illustration of a portion of a coated abrasive with abrasive particles **202** and **203**.

In an embodiment, the abrasive particles may have a random rotational orientation relative to each other. The randomness of the rotational orientation is evaluated by creating a histogram or distribution of measured orientations from randomly sampled areas from a given abrasive article. The process for measuring the rotational orientation of particles on a substrate is started by obtaining a coated abrasive sample that does not include overlying layers on the particles or cleaning the coated abrasive sample to expose the particles, such that the particles are clearly visible. If a coated abrasive article includes layers overlying the particles (e.g., size coat, supersize coat, etc.) a gentle sandblasting operation can be conducted to selectively remove the overlying layers and expose the underlying abrasive particles. Care should be taken during the sandblasting operation to ensure that the particles are not damaged or moved. The selective removal operation may be conducted in stages to ensure that only the overlying layers are removed but the underlying particles are not damaged or altered.

After obtaining a sample with the particles exposed, at least two randomly selected regions of the sample are imaged using a suitable device, such as a Cannon Powershot S110 camera with a resolution of 338 pixels/cm. From these

images, the location and orientation of each particle relative to the edge of the sample are cataloged using MATLAB image analysis software. The orientation of the particle is based on the angle of the major axis of the abrasive particles as viewed top-down relative to an edge of the coated abrasive. The same axis should be used to evaluate all sample images. The orientation of each particle is defined by an orientation angle between -90 degrees and $+90$ degrees. The orientation angles are then plotted in a plot of orientation angle (x-axis) versus frequency (y-axis) to create a histogram of the orientation angles. If the histogram has an essentially flat profile, such that the frequency for any given orientation angle is nearly the same as the frequency for any other orientation angle, the histogram demonstrates that the particles generally have no primary orientation mode, and therefore, the particles have a random orientation. FIG. 3 includes an exemplary image of a portion of an abrasive article having abrasive particles in a random orientation.

It should be noted that while certain embodiments herein can have particles arranged in a random orientation, other embodiments may include particles arranged in a non-random or controlled distribution.

According to one embodiment, an abrasive particle **202** can be overlying the backing **201** in a first position having a first rotational orientation relative to a lateral axis **281** defining the width of the backing **201** and perpendicular to a longitudinal axis **280**. In particular, the abrasive particle **202** can have a predetermined rotational orientation defined by a first rotational angle between a lateral axis **284** parallel to the lateral axis **281** and a dimension of the abrasive particle **202**. Notably, reference herein to a dimension can be a reference to a bisecting axis **231** of the abrasive particle **202** extending through a center point **221** of the abrasive particle **202** as viewed top-down. Moreover, the predetermined rotational orientation can be defined as the smallest angle **241** with the lateral axis **284** extending through the center point **221**. As illustrated in FIG. 2A, the abrasive particle **202** can have a predetermined rotational angle defined as the smallest angle **241** between the bisecting axis **231** and the lateral axis **284**, wherein the lateral axis is parallel to the lateral axis **281**. It will be appreciated that the lateral axis **281** may also be a radial axis where the backing **201** has a circular or elliptical shape. In accordance with an embodiment, the angle **241** defining the rotational orientation of the abrasive particle **202** relative to the lateral axis **284** can be any value within a range between at least 0 degrees and not greater than 90 degrees.

As further illustrated in FIG. 2A, the abrasive particle **203** can be at a second position overlying the backing **201** and having a predetermined rotational orientation. Notably, the predetermined rotational orientation of the abrasive particle **203** can be characterized as the smallest angle between the lateral axis **285** parallel to the lateral axis **281** of the backing and a bisecting axis **232** of the abrasive particle **203** extending through a center point **222** of the abrasive particle **203**. In accordance with an embodiment, the rotational angle **208** can be any value within a range of at least 0 degrees to 90 degrees.

In accordance with an embodiment, the abrasive particle **202** can have a predetermined rotational orientation as defined by the rotational angle **241** that is different than the predetermined rotational orientation of the abrasive particle **203** as defined by the rotational angle **208**. In particular, the difference between the rotational angle **241** and rotational angle **208** for the abrasive particles **202** and **203** can define a predetermined rotational orientation difference. In particular instances, the predetermined rotational orientation dif-

ference can be any value within a range of at least 0 degrees and not greater than 90 degrees.

FIG. 2B includes a top-view illustration of a portion of a coated abrasive article according to an embodiment. As illustrated, the abrasive article **200** can include a plurality of abrasive particles arranged at different positions on the backing **201**, wherein the abrasive particles **253** define a random distribution of the particles on the backing. Moreover, the abrasive particles **253** have a random rotational orientation with respect to each other, such that the rotational orientation of the abrasive particles **253** varies from particle-to-particle in a random manner. According to one aspect, the random rotational orientation of the abrasive particles is such that the rotational angle of one abrasive particle in the group cannot be used to predict the rotational orientation of any of the immediately adjacent particles. Thus, a group of abrasive particles having a random rotational orientation lack any short-range (i.e., immediately adjacent) or long-range order with respect to their rotational angles. It will be appreciated that any particles attached to the backing using the systems and processes of the embodiments herein can have a random rotational orientation with respect to each other.

The coated abrasive articles of the embodiments herein can have at least a majority of the total content (weight or number) of abrasive particles having a random rotational orientation on the backing. In still other instances, at least 10% of the total number of shaped abrasive particles or at least 20% or at least 30% or at least 40% or at least 50% or at least 60% or at least 70% or at least 80% or at least 90% or essentially all of the shaped abrasive particles have a random rotational orientation. In one embodiment, all of the abrasive particles on the backing have a random rotational orientation.

FIG. 4A includes a side-view illustration of abrasive particles on a backing according to an embodiment. The methods disclosed in the embodiments herein can facilitate the formation of coated abrasive articles having a particular distribution and orientation of abrasive particles. Notably, without wishing to be tied to a particular theory, it is noted that the projection rate and efficiency of the process disclosed herein may facilitate improved control of the tilt angle of the abrasive particles adhered to the backing. To better understand these features, FIG. 4A provides a side-view illustration of three abrasive particles in various orientations. It will be appreciated that the coated abrasive articles of the embodiments herein can have various contents of particles in the depicted orientations as described in more detail herein. The first particle **402** can have a particle axis **403** extending at a particular tilt angle **404** relative to the surface of the backing **401**. The particle axis **403** can be parallel to the longitudinal axis of the first particle **402** that defines the length of the first particle **402**. The first particle **402** is representative of a particle in a standing orientation having a tilt angle **404** within a range of greater than 65 degrees to 90 degrees. The second particle **411** can have a particle axis **412** extending at a particular tilt angle **413** relative to the surface of the backing **401**. The particle axis **412** can be parallel to a longitudinal axis of the second particle **411** that defines the length of the second particle **411**. The second particle **411** is representative of a particle in a slanted orientation having a tilt angle **413** within a range of greater than 5 degrees to 65 degrees. The third particle **421** can have a particle axis **422** extending at a particular tilt angle **423** relative to the surface of the backing **401**. The particle axis **422** can be parallel to a longitudinal axis of the third particle **421** that defines the length of the third particle

421. The third particle 421 is representative of a particle in a flat orientation having a tilt angle 423 within a range of 0 degrees to not greater than 5 degrees (i.e., not greater than 5 degrees). FIG. 4B includes a side-view illustration of a particle on a backing having a particular tilt angle according to an embodiment. As illustrated, the particle 431 can be a shaped abrasive particle as described in embodiments herein. The particle 431 can have a longitudinal axis 436 as defined later in this application. The backing 433 can define a substantially planar surface and have an axis 434 extending normal to the substantially planar surface of the backing 433. The tilt angle 435 is the smallest angle between the planar surface of the backing 433 and an axis 432, which extends parallel to the longitudinal axis 436 of the particle 431. Certain particles can have longitudinal axes along various surfaces, which may result in different tilt angles. In such instances, the axis defining the largest angle is the tilt angle.

FIG. 4C includes a top-down illustration of the particle of FIG. 4B. In certain instances, a top-down view may provide a suitable vantage for identifying the direction of the tilt and thus can be suitable for measuring the tilt angle.

FIG. 4D includes a side-view illustration of a particle on a backing having a particular tilt angle according to an embodiment. As illustrated, the particle 441 can have a longitudinal axis 446 as defined later in this application. The particle 441 can be an abrasive particle, and more particularly, can be a non-shaped abrasive particle. The backing 443 can define a substantially planar surface and have an axis 444 extending normal to the substantially planar surface of the backing 443. The tilt angle 445 can be the smallest angle between an axis 442, which extends parallel to the longitudinal axis 446 and the surface of the backing 443. It will be appreciated that certain particles, such as equiaxed particles, will not have a tilt angle.

FIG. 4E includes a top-down illustration of the particle of FIG. 4D. The top-down view may be used to evaluate the tilt angle of the particle. As depicted, the top-down view may be the best view for evaluating the tilt angle as a side-view may not necessarily ensure the smallest angle is identified. A combination of top-down and side-view illustrations may be suitable for identifying and evaluating the tilt angle 445.

In one aspect, a coated abrasive article may include a plurality of abrasive particles, wherein the tilt angle of the abrasive particles is controlled, which may facilitate improved performance of the coated abrasive. For example, at least a portion of the shaped abrasive particles have a tilt angle greater than 45 degrees. In further aspects, a portion includes at least 10% of the total number of shaped abrasive particles or at least 20% or at least 30% or at least 40% or at least 50% or at least 60% or at least 70% or at least 80% or at least 90% or essentially all the shaped abrasive particles have a tilt angle greater than 45 degrees.

In an embodiment, the coated abrasive article may have a particular percentage of standing particles that may facilitate improved performance and/or manufacturing of the abrasive article. Standing particles can be defined as particles having a tilt angle of 65 to 90 degrees. In an embodiment, the standing abrasive particles can include at least 10% of the total number of the abrasive particles or at least 20% or at least 30% or at least 40% or at least 50% or at least 55% or at least 57% or at least 60% or at least 62% or at least 65% or at least 67% or at least 70% or at least 72% or at least 75% or at least 77% or at least 80% or at least 82% or at least 85% or at least 87% or at least 90% of the total number of the abrasive particles. In another embodiment, the standing abrasive particles can include not greater than 99.9% of the

total number of the abrasive particles or not greater than 99% or not greater than 98% or not greater than 97% or not greater than 96% or not greater than 95% of the total number of the abrasive particles. It will be appreciated that the percentage of standing particles can be between any of the minimum and maximum values noted above, including, for example, but not limited to, at least 20% and not greater than 99% or at least 50% and not greater than 95%.

In an embodiment, the coated abrasive article may have a particular percentage of slanted particles that may facilitate improved performance and/or manufacturing of the abrasive article. Slanted particles can be defined as particles having a tilt angle of 5 to 65 degrees. In an embodiment, the slanted abrasive particles can include at least 1% of the total number of the abrasive particles or at least 2% or at least 3% or at least 4% or at least 5% or at least 6% or at least 7% or at least 8% or at least 9% or at least 10% or at least 11% or at least 12% or at least 13% or at least 14% or at least 15% or at least 16% or at least 17% or at least 18% or at least 20% or at least 25% of the total number of the abrasive particles. In another embodiment, the slanted abrasive particles can include not greater than 90% of the total number of the abrasive particles or not greater than 85% or not greater than 80% or not greater than 75% or not greater than 70% or not greater than 65% or not greater than 60% or not greater than 55% or not greater than 50% or not greater than 45% or not greater than 40% or not greater than 35% or not greater than 30% or not greater than 25% or not greater than 20% or not greater than 18% or not greater than 15% or not greater than 12% or not greater than 10% of the total number of the abrasive particles. It will be appreciated that the percentage of slanted particles can be between any of the minimum and maximum values noted above, including, for example, but not limited to, at least 5% and not greater than 80% or at least 15% and not greater than 35%.

In an embodiment, the coated abrasive article may have a particular percentage of well oriented particles that may facilitate improved performance and/or manufacturing of the abrasive article. Well oriented particles can be defined as particles having a tilt angle of 5 to 90 degrees and include slanted and standing particles. In an embodiment, the well oriented abrasive particles can include at least 60% of the total number of the abrasive particles or at least 62% or at least 65% or at least 67% or at least 70% or at least 72% or at least 75% or at least 77% or at least 80% or at least 82% or at least 85% or at least 87% or at least 90% or at least 92% or at least 95% of the total number of the abrasive particles. In another embodiment, the well oriented abrasive particles can be not greater than 99.9% of the total number of the abrasive particles or not greater than 99% or not greater than 98% or not greater than 97% or not greater than 96% or not greater than 95%. It will be appreciated that the percentage of well oriented particles can be between any of the minimum and maximum values noted above, including, for example, but not limited to, at least 5% and not greater than 80% or at least 15% and not greater than 35%.

In an embodiment, the coated abrasive article may have a particular percentage of fallen particles that may facilitate improved performance and/or manufacturing of the abrasive article. Fallen particles can be defined as particles having a tilt angle of 0 to 5 degrees. In an embodiment, the fallen abrasive particles at least 0.1% of the total number of the abrasive particles or at least 0.2% or at least 0.4% or at least 0.6% or at least 0.8% or at least 1% or at least 1.5% or at least 2% or at least 2.5% or at least 3% or at least 3.5% or at least 4% or at least 4.5% or at least 5% of the total number of the abrasive particles. In another embodiment, the fallen

abrasive particles can include not greater than 20% of the total number of the abrasive particles or not greater than 18% or not greater than 15% or not greater than 12% or not greater than 10% or not greater than 9% or not greater than 8% or not greater than 7% or not greater than 6% or not greater than 5% of the total number of the abrasive particles. It will be appreciated that the percentage of fallen particles can be between any of the minimum and maximum values noted above, including, for example, but not limited to, at least 0.2% and not greater than 15% or at least 1% and not greater than 9%.

In an embodiment, the coated abrasive article may have a particular percentage of inverted particles that may facilitate improved performance and/or manufacturing of the abrasive article. Inverted particles can be defined as particles having a tilt angle of 5 to 90 degrees as well as having a tip, corner, or point extending into the make coat, and a planar surface or surfaces such as a base, opposite the tip on the other end of the abrasive particle. Only particles having a tip on one end of its longitudinal axis and at least one planar surface on the opposite end of the longitudinal axis can be inverted. Exemplary particle shapes that can be in an inverted orientation include triangles, 3-PT (3-PT) stars, pentagons, and pyramids. Particles having planar surfaces on both ends of their longitudinal axis (e.g., rods or cylinders, rectangular prisms,) and particles having points on both ends of their longitudinal axis [e.g., toothpick-shaped, diamond-shaped, 4-pointed (4-PT) stars] cannot be in an inverted orientation. Inverted particles are not standing, slanted, fallen, or well oriented. In an embodiment, the inverted abrasive particles make up at least 0.1% of the total number of the abrasive particles or at least 0.2% or at least 0.4% or at least 0.6% or at least 0.8% or at least 1% or at least 1.5% or at least 2% or at least 2.5% or at least 3% or at least 3.5% or at least 4% or at least 4.5% or at least 5% of the total number of the abrasive particles. In another embodiment, the inverted abrasive particles can include not greater than 20% of the total number of the abrasive particles or not greater than 18% or not greater than 15% or not greater than 12% or not greater than 10% or not greater than 9% or not greater than 8% or not greater than 7% or not greater than 6% or not greater than 5% of the total number of the abrasive particles. It will be appreciated that the percentage of inverted particles can be between any of the minimum and maximum values noted above, including, for example, but not limited to, at least 0.2% and not greater than 15% or at least 1% and not greater than 9%.

In an embodiment, the coated abrasive particle may have a particular ratio (Pst/Psl) of standing particles (Pst) to slanted particles (Psl) that may facilitate improved performance and/or manufacturing of the abrasive article. In an embodiment, Pst/Psl can be at least 1 or at least 1.2 or at least 1.4 or at least 1.6 or at least 1.8 or at least 2.0 or at least 2.2 or at least 2.4 or at least 2.6 or at least 2.8 or at least 3.0 or at least 3.2 or at least 3.4 or at least 3.6 or at least 3.8 or at least 4.0 or at least 4.2 or at least 4.4 or at least 4.6 or at least 4.8 or at least 5.0 or at least 5.2 or at least 5.4 or at least 5.6. In another embodiment, Pst/Psl can be not greater than 100 or not greater than 95 or not greater than 90 or not greater than 80 or not greater than 70 or not greater than 60 or not greater than 50 or not greater than 40 or not greater than 30 or not greater than 20 or not greater than 10. It will be appreciated that Pst/Psl can be between any of the minimum and maximum values noted above, including, for example, but not limited to, at least 1.2 and not greater than 95 or at least 2.0 and not greater than 40.

In an embodiment, the coated abrasive particle may have a particular ratio (Pst/Pf) of standing particles (Pst) to fallen particles (Pf) that may facilitate improved performance and/or manufacturing of the abrasive article. In an embodiment, Pst/Pf can be at least 2.0 or at least 2.2 or at least 2.4 or at least 2.6 or at least 2.8 or at least 3.0 or at least 3.2 or at least 3.4 or at least 3.6 or at least 3.8 or at least 4.0 or at least 4.2 or at least 4.4 or at least 4.6 or at least 4.8 or at least 5.0 or at least 5.2 or at least 5.4 or at least 5.6 or at least 5.8 or at least 6.0 or at least 6.2 or at least 6.4 or at least 6.6 or at least 6.8 or at least 7.0 or at least 7.2 or at least 7.4 or at least 7.6 or at least 7.8 or at least 8.0. In another embodiment, Pst/Pf can be not greater than 1000 or not greater than 800 or not greater than 500 or not greater than 200 or not greater than 100 or not greater than 95. It will be appreciated that Pst/Pf can be between any of the minimum and maximum values noted above, including, for example, but not limited to, at least 2.0 and not greater than 500 or at least 2.6 and not greater than 95.

In an embodiment, the coated abrasive particle may have a particular ratio (Psl/Pt) of slanted particles (Psl) to fallen particles (Pt) that may facilitate improved performance and/or manufacturing of the abrasive article. In an embodiment, Psl/Pt can be at least 1 or at least 1.2 or at least 1.4 or at least 1.6 or at least 1.8 or at least 2.0 or at least 2.2 or at least 2.4 or at least 2.6 or at least 2.8 or at least 3.0 or at least 3.2 or at least 3.4 or at least 3.6 or at least 3.8 or at least 4.0 or at least 4.2 or at least 4.4 or at least 4.6 or at least 4.8 or at least 5.0 or at least 5.2 or at least 5.4 or at least 5.6. In another embodiment, Psl/Pt can be not greater than 100 or not greater than 95 or not greater than 90 or not greater than 80 or not greater than 70 or not greater than 60 or not greater than 50 or not greater than 40 or not greater than 30 or not greater than 20 or not greater than 10 or not greater than 8 or not greater than 6. It will be appreciated that Psl/Pt can be between any of the minimum and maximum values noted above, including, for example, but not limited to, at least 2.0 and not greater than 95 or at least 2.6 and not greater than 70.

In an embodiment, the coated abrasive particle may have a particular ratio (Pst/Pi) of standing particles (Pst) to inverted particles (Pi) that may facilitate improved performance and/or manufacturing of the abrasive article. In an embodiment, Pst/Pi can be at least 1 or at least 2.0 or at least 3.0 or at least 4.0 or at least 5.0 or at least 6 or at least 7 or at least 8 or at least 9 or at least 10 or at least 12 or at least 15 or at least 18 or at least 20 or at least 25 or at least 30 or at least 40 or at least 50. In another embodiment, Pst/Pi can be not greater than 100 or not greater than 95 or not greater than 90 or not greater than 80 or not greater than 70 or not greater than 60 or not greater than 50 or not greater than 40 or not greater than 30 or not greater than 20 or not greater than 10 or not greater than 8 or not greater than 6. It will be appreciated that Pst/Pi can be between any of the minimum and maximum values noted above, including, for example, but not limited to, at least 2.0 and not greater than 80 or at least 6 and not greater than 20.

In an embodiment, the coated abrasive particle may have a particular ratio (Psl/Pi) of slanted particles (Psl) to inverted particles (Pi) that may facilitate improved performance and/or manufacturing of the abrasive article. In an embodiment, Psl/Pi can be at least 0.6 or at least 0.7 or at least 0.8 or at least 0.9 or at least 1 or at least 1.2 or at least 1.4 or at least 1.6 or at least 1.8 or at least 2.0 or at least 2.2 or at least 2.4 or at least 2.6 or at least 2.8 or at least 3.0 or at least 3.2 or at least 3.4 or at least 3.6 or at least 3.8 or at least 4.0 or at least 4.2 or at least 4.4 or at least 4.6 or at least

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4.8 or at least 5.0 or at least 5.2 or at least 5.4 or at least 5.6. In another embodiment, Psl/Pi can be not greater than 100 or not greater than 95 or not greater than 90 or not greater than 80 or not greater than 70 or not greater than 60 or not greater than 50 or not greater than 40 or not greater than 30 or not greater than 20 or not greater than 10 or not greater than 8 or not greater than 6 or not greater than 4 or not greater than 3 or not greater than 2 or not greater than 1.5. It will be appreciated that Psl/Pi can be between any of the minimum and maximum values noted above, including, for example, but not limited to, at least 2.0 and not greater than 95 or at least 2.6 and not greater than 70.

In an embodiment, the coated abrasive particle may have a particular ratio (Pf/Pi) of fallen particles (Pt) to inverted particles (Pi) that may facilitate improved performance and/or manufacturing of the abrasive article. In an embodiment, Pf/Pi can be at least 0.6 or at least 0.7 or at least 0.8 or at least 0.9 or at least 1 or at least 1.2 or at least 1.4 or at least 1.6 or at least 1.8 or at least 2.0 or at least 2.2 or at least 2.4 or at least 2.6 or at least 2.8 or at least 3.0 or at least 3.2 or at least 3.4 or at least 3.6 or at least 3.8 or at least 4.0 or at least 4.2 or at least 4.4 or at least 4.6 or at least 4.8 or at least 5.0. In another embodiment, Pf/Pi can be not greater than 100 or not greater than 95 or not greater than 90 or not greater than 80 or not greater than 70 or not greater than 60 or not greater than 50 or not greater than 40 or not greater than 30 or not greater than 20 or not greater than 10 or not greater than 8 or not greater than 6 or not greater than 4 or not greater than 3 or not greater than 2 or not greater than 1.5. It will be appreciated that Pf/Pi can be between any of the minimum and maximum values noted above, including, for example, but not limited to, at least 2.0 and not greater than 95 or at least 2.6 and not greater than 70.

Embodiments herein have referred to particles, which can include abrasive particles, secondary particles, or any combination thereof. Various types of abrasive particles and/or secondary particles can be used with abrasive articles described in the embodiments herein. FIG. 5A includes a perspective view illustration of a shaped abrasive particle in accordance with an embodiment. The shaped abrasive particle 500 can include a body 501 including a major surface 502, a major surface 503, and a side surface 504 extending between the major surfaces 502 and 503. As illustrated in FIG. 5A, the body 501 of the shaped abrasive particle 500 can be a thin-shaped body, wherein the major surfaces 502 and 503 are larger than the side surface 504. Moreover, the body 501 can include a longitudinal axis 510 extending from a point to a base and through the midpoint 550 on a major surface 502 or 503. The longitudinal axis 510 can define the longest dimension of the body along a major surface and through the midpoint 550 of the major surface 502.

In certain particles, if the midpoint of a major surface of the body is not readily apparent, one may view the major surface top-down, draw a closest-fit circle around the two-dimensional shape of the major surface and use the center of the circle as the midpoint of the major surface. FIG. 5B includes a top-down illustration of the shaped abrasive particle of FIG. 5A. Notably, the body 501 includes a major surface 502 having a triangular two-dimensional shape. The circle 560 is drawn around the triangular shape to facilitate the location of the midpoint 550 on the major surface 502.

Referring again to FIG. 5A, the body 501 can further include a lateral axis 511 defining a width of the body 501 extending generally perpendicular to the longitudinal axis 510 on the same major surface 502. Finally, as illustrated, the body 501 can include a vertical axis 512, which in the context of thin-shaped bodies can define a height (or thick-

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ness) of the body 501. For thin-shaped bodies, the length of the longitudinal axis 510 is greater than the vertical axis 512. As illustrated, the thickness 512 can extend along the side surface 504 between the major surfaces 502 and 503 and perpendicular to the plane defined by the longitudinal axis 510 and lateral axis 511. It will be appreciated that reference herein to length, width, and height of the abrasive particles may be reference to average values taken from a suitable sampling size of abrasive particles of a larger group, including, for example, a group of abrasive particles affixed to a fixed abrasive.

FIG. 5A includes an illustration of a shaped abrasive particle having a two-dimensional shape as defined by the plane of the upper major surface 502 or major surface 503, which has a generally triangular two-dimensional shape. It will be appreciated that the shaped abrasive particles of the embodiments herein are not so limited and can include other two-dimensional shapes. For example, the shaped abrasive particles of the embodiment herein can include particles having a body with a two-dimensional shape as defined by a major surface of the body from the group of shapes including polygons, regular polygons, irregular polygons, irregular polygons including arcuate or curved sides or portions of sides, ellipsoids, numerals, Greek alphabet characters, Latin alphabet characters, Russian alphabet characters, Kanji characters, complex shapes having a combination of polygons shapes, shapes including a central region and a plurality of arms (e.g., at least three arms) extending from a central region (e.g., star shapes), and a combination thereof. Particular polygonal shapes include rectangular, trapezoidal, quadrilateral, pentagonal, hexagonal, heptagonal, octagonal, nonagonal, decagonal, and any combination thereof. In another instance, the finally formed shaped abrasive particles can have a body having a two-dimensional shape such as an irregular quadrilateral, an irregular rectangle, an irregular trapezoid, an irregular pentagon, an irregular hexagon, an irregular heptagon, an irregular octagon, an irregular nonagon, an irregular decagon, and a combination thereof. An irregular polygonal shape is one where at least one of the sides defining the polygonal shape is different in dimension (e.g., length) with respect to another side. As illustrated in other embodiments herein, the two-dimensional shape of certain shaped abrasive particles can have a particular number of exterior points or external corners. For example, the body of the shaped abrasive particles can have a two-dimensional polygonal shape as viewed in a plane defined by a length and width, wherein the body comprises a two-dimensional shape having at least 4 exterior points (e.g., a quadrilateral), at least 5 exterior points (e.g., a pentagon), at least 6 exterior points (e.g., a hexagon), at least 7 exterior points (e.g., a heptagon), at least 8 exterior points (e.g., an octagon), at least 9 exterior points (e.g., a nonagon), and the like.

FIG. 7 includes a perspective view illustration of a shaped abrasive particle according to another embodiment. Notably, the shaped abrasive particle 700 can include a body 701 including a surface 702 and a surface 703, which may be referred to as end surfaces 702 and 703. The body can further include major surfaces 704, 705, 706, 707 extending between and coupled to the end surfaces 702 and 703. The shaped abrasive particle of FIG. 7 is an elongated shaped abrasive particle having a longitudinal axis 710 that extends along the major surface 705 and through the midpoint 740 between the end surfaces 702 and 703. For particles having an identifiable two-dimensional shape, such as the shaped abrasive particles of FIGS. 5 and 7, the longitudinal axis is the dimension that would be readily understood to define the

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length of the body through the midpoint on a major surface. For example, in FIG. 7, the longitudinal axis **710** of the shaped abrasive particle **700** extends between the end surfaces **702** and **703** parallel to the edges defining the major surface as shown. Such a longitudinal axis is consistent with how one would define the length of a rod. Notably, the longitudinal axis **710** does not extend diagonally between the corners joining the end surfaces **702** and **703** and the edges defining the major surface **705**, even though such a line may define the dimension of greatest length. To the extent that a major surface has undulations or minor imperfections from a perfectly planar surface, the longitudinal axis can be determined using a top-down, two-dimensional image that ignores the undulations.

It will be appreciated that the surface **705** is selected for illustrating the longitudinal axis **710** because the body **701** has a generally square cross-sectional contour as defined by the end surfaces **702** and **703**. As such, the surfaces **704**, **705**, **706**, and **707** can be approximately the same size relative to each other. However, in the context of other elongated abrasive particles, the surfaces **702** and **703** can have a different shape, for example, a rectangular shape, and as such, at least one of the surfaces **704**, **705**, **706**, and **707** may be larger relative to the others. In such instances, the largest surface can define the major surface and the longitudinal axis would extend along the largest of those surfaces through the midpoint **740** and may extend parallel to the edges defining the major surface. As further illustrated, the body **701** can include a lateral axis **711** extending perpendicular to the longitudinal axis **710** within the same plane defined by the surface **705**. As further illustrated, the body **701** can further include a vertical axis **712** defining a height of the abrasive particle, wherein the vertical axis **712** extends in a direction perpendicular to the plane defined by the longitudinal axis **710** and lateral axis **711** of the surface **705**.

It will be appreciated that like the thin-shaped abrasive particle of FIG. 5, the elongated shaped abrasive particle of FIG. 7 can have various two-dimensional shapes, such as those defined with respect to the shaped abrasive particle of FIG. 5. The two-dimensional shape of the body **701** can be defined by the shape of the perimeter of the end surfaces **702** and **703**. The elongated shaped abrasive particle **700** can have any of the attributes of the shaped abrasive particles of the embodiments herein.

FIG. 8A includes a perspective view illustration of a controlled height abrasive particle according (CHAP) to an embodiment. As illustrated, the CHAP **800** can include a body **801** including a first major surface **802**, a second major surface **803**, and a side surface **804** extending between the first and second major surfaces **802** and **803**. As illustrated in FIG. 8A, the body **801** can have a thin, relatively planar shape, wherein the first and second major surfaces **802** and **803** are larger than the side surface **804** and substantially parallel to each other. Moreover, the body **801** can include a longitudinal axis **810** extending through the midpoint **820** and defining a length of the body **801**. The body **801** can further include a lateral axis **811** on the first major surface **802**, which extends through the midpoint **820** of the first major surface **802**, perpendicular to the longitudinal axis **810**, and defining a width of the body **801**.

The body **801** can further include a vertical axis **812**, which can define a height (or thickness) of the body **801**. As illustrated, the vertical axis **812** can extend along the side surface **804** between the first and second major surfaces **802** and **803** in a direction generally perpendicular to the plane defined by the axes **810** and **811** on the first major surface. For thin-shaped bodies, such as the CHAP illustrated in FIG.

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8A, the length can be equal to or greater than the width and the length can be greater than the height. It will be appreciated that reference herein to length, width, and height of the abrasive particles may be referenced to average values taken from a suitable sampling size of abrasive particles of a batch of abrasive particles.

Unlike the shaped abrasive particles of FIGS. 5A, 5B, and 7, the CHAP of FIG. 8A does not have a readily identifiable two-dimensional shape based on the perimeter of the first or second major surfaces **802** and **803**. Such abrasive particles may be formed in a variety of ways, including but not limited to, fracturing of a thin layer of material to form abrasive particles having a controlled height but with irregularly formed, planar, major surfaces. For such particles, the longitudinal axis is defined as the longest dimension on the major surface that extends through a midpoint on the surface. To the extent that the major surface has undulations, the longitudinal axis can be determined using a top-down, two-dimensional image that ignores the undulations. Moreover, as noted above in FIG. 5B, a closest-fit circle may be used to identify the midpoint of the major surface and identification of the longitudinal and lateral axes.

FIG. 8B includes an illustration of a non-shaped particle, which may be an elongated, non-shaped abrasive particle or a secondary particle, such as a diluent grain, a filler, an agglomerate, or the like. Shaped abrasive particles may be formed through particular processes, including molding, printing, casting, extrusion, and the like. Shaped abrasive particles can be formed such that each particle has substantially the same arrangement of surfaces and edges relative to each other. For example, a group of shaped abrasive particles generally have the same arrangement and orientation and or two-dimensional shape of the surfaces and edges relative to each other. As such, the shaped abrasive particles have a relatively high shape fidelity and consistency in the arrangement of the surfaces and edges relative to each other. Moreover, constant height abrasive particles (CHAPs) can also be formed through particular processes that facilitate the formation of thin-shaped bodies that can have irregular two-dimensional shapes when viewing the major surface top-down. CHAPs can have less shape fidelity than shaped abrasive particles but can have substantially planar and parallel major surfaces separated by a side surface.

By contrast, non-shaped particles can be formed through different processes and have different shape attributes compared to shaped abrasive particles and CHAPs. For example, non-shaped particles are typically formed by a comminution process wherein a mass of material is formed and then crushed and sieved to obtain abrasive particles of a certain size. However, a non-shaped particle will have a generally random arrangement of surfaces and edges, and generally will lack any recognizable two-dimensional or three-dimensional shape in the arrangement of the surfaces and edges. Moreover, non-shaped particles do not necessarily have a consistent shape with respect to each other, and therefore have a significantly lower shape fidelity compared to shaped abrasive particles or CHAPs. The non-shaped particles generally are defined by a random arrangement of surfaces and edges for each particle and with respect to other non-shaped particles.

FIG. 8B includes a perspective view illustration of a non-shaped particle. The non-shaped particle **850** can have a body **851** including a generally random arrangement of edges **855** extending along the exterior surface of the body **851**. The body can further include a longitudinal axis **852** defining the longest dimension of the particle. The longitudinal axis **852** defines the longest dimension of the body as

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viewed in two-dimensions. Thus, unlike shaped abrasive particles and CHAPs, where the longitudinal axis is measured on the major surface, the longitudinal axis of a non-shaped particle is defined by the points on the body furthest from each other as the particle is viewed in two-dimensions using an image or vantage that provides a view of the particle's longest dimension. That is, an elongated particle, but non-shaped particles, such as illustrated in FIG. 8B, should be viewed in a perspective that makes the longest dimension apparent to properly evaluate the longitudinal axis. The body **851** can further include a lateral axis **853** extending perpendicular to the longitudinal axis **852** and defining a width of the particle. The lateral axis **853** can extend perpendicular to the longitudinal axis **852** through the midpoint **856** of the longitudinal axis in the same plane used to identify the longitudinal axis **852**. The abrasive particle may have a height (or thickness) as defined by the vertical axis **854**. The vertical axis **854** can extend through the midpoint **856** but in a direction perpendicular to the plane used to define the longitudinal axis **852** and lateral axis **853**. To evaluate the height, one may have to change the perspective of view of the abrasive particle to look at the particle from a different vantage than is used to evaluate the length and width.

In an embodiment, the plurality of abrasive particles **102** and **103** of the coated abrasive article can include shaped abrasive particles. In an embodiment, the shaped abrasive particles can be 3-PT star-shaped abrasive particles. The abrasive particles can have a length (l), a width (w), and a thickness (t), wherein the width $\geq$ thickness and the length $\geq$ thickness. The particles can have a primary aspect ratio based on the length:width of the body. The particles can have a secondary aspect ratio based on the length:thickness of the body. The particles can also have a tertiary aspect ratio, based on the width:thickness of the body. The particles **102** and **103** can be an elongated abrasive particle, having a primary aspect ratio greater than 1.1:1.

In an embodiment, the plurality of shaped abrasive particles can include a plurality of shaped abrasive particles having a 3-PT star two-dimensional shape as viewed in a plane of a length and width of the body. The body can include at least 3 exterior corners and at least 4 side surface sections, or at least 5 side surface sections or at least 6 side surface sections. In an embodiment, the plurality of shaped abrasive particles can include a body having at least 3 exterior corners, where the sum of the angles of the exterior corners is less than 180 degrees. In an embodiment, the plurality of shaped abrasive particles can include a body having at least 3 exterior corners, where each of the exterior corners defines an angle less than 60 degrees or less than 59 degrees or less than 58 degrees or less than 57 degrees or less than 56 degrees or less than 55 degrees. In an embodiment, the plurality of shaped abrasive particles can include comprises a body having at least 3 exterior corners and at least 3 interior corners, where each of the interior corners have an interior corner angle value greater than any of the exterior corner values of any of the at least 3 exterior corners.

Exterior corners can be identified using the "rubber band test". If a rubber band were to be stretched around the body of the abrasive particle, the corners that contact the rubber band and cause deflection of the rubber band would be exterior corners.

FIG. 6A includes a top view image of a 3-PT star-shaped abrasive particle according to a particular embodiment. As illustrated, the shaped abrasive particle **600** can define a star-shaped body, as viewed in two dimensions. In particular, the shaped abrasive particle **600** can include a body **601**

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having a central portion **602** and a first arm **603**, a second arm **604**, and a third arm **605** extending from the central portion **602**. The body **601** can have a length (l) measured as the longest dimension along a side of the particle and a width (w), measured as the longest dimension of the particle between a midpoint **653** of a side through the midpoint **690** of the body **601** to a first tip **606** of the first arm **603**. The width can extend in a direction perpendicular to the dimension of the length. The body **601** can have a thickness (t), extending in a direction perpendicular to the upper surface or first major surface **610** of the body **601** defining the third side surface **656** between the upper surface or first major surface **610** and the base surface **611** as illustrated in FIG. 6B, which is a side view illustration of the image of the particle of FIG. 6A.

The shaped abrasive particle **600** can have a body **601** in the form of a 3-PT star defined by the first arm **603**, second arm **604**, and the third arm **605** extending from the central portion **602**. According to one particular embodiment, at least one of the arms, including, for example, the first arm **603**, can have a midpoint width **613** that is less than a central portion width **612**. The central portion **602** can be defined as a region between the midpoints **651**, **652**, and **653** of the first side surface **654**, second side surface **655**, and third side surface **656**, respectively. The central portion width **612** of the first arm **603** can be the width of the dimension between the midpoints **651** and **652**. The midpoint width **613** can be the width of the line at a midpoint between the line of the central portion width **612** and the tip **606** of the first arm **603** along a first axis **660**. In certain instances, the midpoint width **613** can be not greater than about 90% of the central portion width **612**, such as not greater than about 80%, not greater than about 70%, not greater than about 5%, or even not greater than about 60%. Still, the midpoint width **613** can be at least about 10%, such as at least about 20%, at least about 30%, or even at least about 40% of the central portion width **612**. It will be appreciated that the midpoint width **613** can have a width relative to the central portion width **612** within a range between any of the above minimum and maximum percentages.

Moreover, the body **601** can have at least one arm, such as the first arm **603**, having a tip width at the tip **606** of the first arm **603** that is less than a midpoint width **613**. In such instances wherein the tip **606** is sharply formed, the tip width may be considered 0. In instances wherein the tip **606** has a radius of curvature, the tip width may be considered the diameter of the circle defined by the radius of curvature. According to one embodiment, the tip width **614** can be not greater than about 90% of the midpoint width **613**, such as not greater than about 80%, not greater than about 70%, not greater than about 60%, not greater than about 50%, not greater than about 40%, not greater than about 30%, not greater than about 20%, or even not greater than about 10%. Still, in certain non-limiting embodiments, the tip width **614** can be at least about 1%, such as at least about 2%, at least about 3%, at least about 5%, or even at least about 10% of the midpoint width **613**. It will be appreciated that the tip width **614** can have a width relative to the midpoint width **613** within a range between any of the above minimum and maximum percentages.

As further illustrated, the body **601** can have a first arm **603** including a first tip **606** defining a first tip angle **621** between the first side surface **654** and the second side surface **655**. According to an embodiment, the first tip angle can be less than about 60 degrees, such as not greater than about 55 degrees, not greater than about 50 degrees, not greater than about 45 degrees, or even not greater than about

40 degrees. Still, the first tip angle can be at least about 5 degrees, such as at least about 8 degrees, at least about 10 degrees, at least about 15 degrees, at least about 20 degrees, at least about 25 degrees, or even at least about 30 degrees. The first tip angle can be within a range between any of the minimum and maximum values noted above.

The body **601** can include a second arm **604** having a second tip **607** defining a second tip angle **622** between the second side surface **655** and third side surface **656**. The second tip angle can be substantially the same as the first tip angle, such as within 5% of the angle numerical value. Alternatively, the second tip angle can be substantially different relative to the first tip angle.

The body **601** can include a third arm **605** having a third tip **608** defining a third tip angle **623** between the first side surface **654** and third side surface **656**. The third tip angle can be substantially the same as the first tip angle or second tip angle, such as within 5% of the angle numerical value. Alternatively, the third tip angle can be substantially different relative to the first tip angle or the second tip angle.

The body **601** can have a total angle, which is a sum of the value of the first tip angle, second tip angle, and third tip angle which can be less than about 180 degrees. In other embodiments, the total angle can be not greater than about 175 degrees, such as not greater than about 170 degrees, not greater than about 15 degrees, not greater than about 150 degrees, such as not greater than about 140 degrees, not greater than about 130 degrees, not greater than about 125 degrees, or even not greater than about 120 degrees. Still, in one non-limiting embodiment, the body **601** can have a total angle of at least about 60 degrees, such as at least about 70 degrees, at least about 80 degrees, at least about 90 degrees, such as at least about 95 degrees, at least about 100 degrees, or even at least about 105 degrees. It will be appreciated that the total sum angle can be within a range between any of the minimum and maximum values noted above.

As noted herein, the body **601** can have a first side surface **654** extending between the first arm **606** and the third arm **608**. In certain instances, the first side surface **654** can have an arcuate contour. For example, turning briefly to FIG. 6C, a top view image of a shaped abrasive particle according to an embodiment is provided. Notably, the shaped abrasive particle of FIG. 6C can include a 3-PT star having a body **681** and an arcuate side surface **682** extending between two points. In particular instances, the side surface **682** can have a concave contour defining a curved portion extending inward toward the central portion **683** of the body **681**.

Referring again to FIG. 6A, the body **601** can have a first side surface **654** having a first side section **658** and a second side section **659**. The first side section **658** can extend between the first tip **606** and the midpoint **651** and the second side section **659** can extend between the third tip **608** and the midpoint **651**. The first side section **658** and second side section **659** can define an interior angle **662** that can be obtuse. For example, the interior angle **662** can be greater than about 90 degrees, such as greater than about 95 degrees, greater than about 100 degrees, greater than about 110 degrees, or even greater than about 120 degrees. Still, in one non-limiting embodiment, the interior angle **662** can be not greater than about 320 degrees, such as not greater than about 300 degrees, or even not greater than about 270 degrees. It will be appreciated that the interior angle can be within a range between any of the minimum and maximum values noted above.

The first side section **658** can extend for a significant portion of the length of the first side surface **654**. For example, the first side section **658** can extend for at least

about 20%, such as at least about 25%, at least about 30%, at least about 35%, or even at least about 40% of a total length of the first side surface **654**. Still, in one non-limiting embodiment, the first side section **658** can have a length (ls1) between the midpoint **651** and the first tip **606** of not greater than about 80%, such as not greater than about 75%, not greater than about 70%, or even not greater than about 5% of the total length of the side surface **654**. It will be appreciated that the length of the first side section **658** can be within a range between any of the minimum and maximum percentages noted above.

The second side section **659** can extend for a significant portion of the length of the first side surface **654**. For example, the second side section **659** can extend for at least about 20%, such as at least about 25%, at least about 30%, at least about 35%, or even at least about 40% of a total length of the first side surface **654**. Still, in one non-limiting embodiment, the second side section **659** can have a length (ls2) between the midpoint **651** and the third tip **608** of not greater than about 80%, such as not greater than about 75%, not greater than about 70%, or even not greater than about 5% of the total length of the side surface **654** as a straight line between the first tip **606** and the third tip **608**. It will be appreciated that the length of the second side section **659** can be within a range between any of the minimum and maximum percentages noted above.

The body **601** can include a first average side surface angle **631** between the side surfaces **654**, **655**, and **656** and the upper surface or first major surface **610**. The body can also include a second side surface angle **632** between the side surfaces **654**, **655**, and **656** and the second major surface or base surface **612**.

In an embodiment, the abrasive particles may include a particular first side surface angle that may facilitate improved performance and/or manufacturing of the abrasive particles. In an embodiment, the first side surface angle can be within a range of at least 70 degrees and not greater than 94 degrees or within a range of at least 80 degrees and not greater than 93 degrees or within a range of at least 83 degrees and not greater than 92 degrees or within a range of at least 85 degrees and not greater than 91 degrees.

In an embodiment, the abrasive particles may include a particular second side surface angle that may facilitate improved performance and/or manufacturing of the abrasive particles. In an embodiment, the second side surface angle can be within a range of at least 70 degrees and not greater than 94 degrees or within a range of at least 80 degrees and not greater than 93 degrees or within a range of at least 83 degrees and not greater than 92 degrees or within a range of at least 85 degrees and not greater than 91 degrees.

While the foregoing body **601** of the 3-PT star has been shown to have an upper surface **610** having a two-dimensional shape, as viewed in the plane of the length and width of the body, that is substantially the same as the two-dimensional shape of the base surface or second major surface **611** of the body **601**, other shapes are contemplated. For example, in one embodiment, the cross-sectional shape of the body at the base surface can define a base surface shape from the group consisting of a 3-PT star, a 4-PT star, a cross-shape, a polygon, ellipsoids, numerals, Greek alphabet characters, Latin alphabet characters, Russian alphabet characters, complex shapes having a combination of polygonal shapes, and a combination thereof. Moreover, the cross-sectional shape of the body at the upper surface can define an upper surface shape, which can be different than the base surface shape and selected from the group of a 3-PT star, a 4-PT star, a cross-shape, a polygon, ellipsoids, numerals,

Greek alphabet characters, Latin alphabet characters, Russian alphabet characters, complex shapes having a combination of polygonal shapes, and a combination thereof.

In particular instances, the upper surface shape can have an arcuate form of the base surface shape. For example, the upper surface shape can define an arcuate 3-PT two-dimensional shape, wherein the arcuate 3-PT two-dimensional shape defines arms having rounded ends. In particular, the arms as defined at the base surface can have a smaller radius of curvature at the tip as compared to the radius of curvature of the corresponding tip at the upper surface.

As described in other embodiments herein, it will be appreciated that at least one of the arms of the body 601 may be formed to have a twist, such that the arm twists around a central axis. For example, the first arm 603 may twist around the axis 660. Moreover, the body 601 can be formed such that at least one arm extends in an arcuate path from the central region.

In an embodiment, the plurality of shaped abrasive particles may define a first group of abrasive particles. In an embodiment, the first group of abrasive particles may include at least two different types of shaped abrasive particles, wherein the two different types of shaped abrasive particles are different from each other based on at least one characteristic selected from the group of particle size, two-dimensional shape, three-dimensional shape, composition, hardness, toughness, friability, density, grain size, agglomeration state, lateral position, longitudinal position, rotational orientation, or any combination thereof.

In an embodiment, the abrasive article may include a second group of abrasive particles different than the first group of abrasive particles. The second group of abrasive particles can be different from the first group of abrasive particles based on at least one characteristic selected from the group of particle size, two-dimensional shape, three-dimensional shape, composition, hardness, toughness, friability, density, grain size, agglomeration state, lateral position, longitudinal position, rotational orientation, or any combination thereof. In a further embodiment, the second group of abrasive particles comprises diluent abrasive particles. In another embodiment, the second group of particles can include randomly shaped or non-shaped abrasive particles.

Shaped abrasive particles may be formed through particular processes, including molding, printing, casting, extrusion, and the like. Shaped abrasive particles can be formed such that each particle has substantially the same arrangement of surfaces and edges relative to each other. For example, a group of shaped abrasive particles generally have the same arrangement and orientation and or two-dimensional shape of the surfaces and edges relative to each other. As such, the shaped abrasive particles have a relatively high shape fidelity and consistency in the arrangement of the surfaces and edges relative to each other. By contrast, non-shaped particles can be formed through different processes and have different shape attributes compared to shaped abrasive particles. For example, non-shaped particles are typically formed by a comminution process wherein a mass of material is formed and then crushed and sieved to obtain abrasive particles of a certain size. However, a non-shaped particle will have a generally random arrangement of surfaces and edges, and generally will lack any recognizable two-dimensional or three-dimensional shape in the arrangement of the surfaces and edges. Moreover, non-shaped particles do not necessarily have a consistent shape with respect to each other, and therefore have a significantly lower shape fidelity compared to shaped abrasive particles.

The non-shaped particles generally are defined by a random arrangement of surfaces and edges for each particle and with respect to other non-shaped particles.

In an embodiment, the abrasive article can include a certain percentage of cracked abrasive particles that may facilitate improved performance or manufacturing of the abrasive article. As defined herein, cracks in the plurality of shaped abrasive particles include cracks visible with a magnification such that the width of the particle is equal to 50% to 95% of the field of view. In an embodiment, the abrasive article can include a plurality of shaped abrasive particles having at least 3 interior corners where not greater than 50% of the total number of shaped abrasive particles have a crack at an interior corner on the first major surface or not greater than 45% or not greater than 40% or not greater than 35% or not greater than 30% or not greater than 25% or not greater than 20% or not greater than 18% or not greater than 16% or not greater than 14% or not greater than 12% or not greater than 10% or not greater than 9% or not greater than 8% or not greater than 7% or not greater than 6% or not greater than 5% or not greater than 4% or not greater than 3% or not greater than 2% or not greater than 1%. In an embodiment, the abrasive article can include a plurality of shaped abrasive particles having at least 3 interior corners where not greater than 50% of the total number of shaped abrasive particles have a crack at an interior corner on the first major surface or not greater than 45% or not greater than 40% or not greater than 35% or not greater than 30% or not greater than 25% or not greater than 20% or not greater than 18% or not greater than 16% or not greater than 14% or not greater than 12% or not greater than 10% or not greater than 9% or not greater than 8% or not greater than 7% or not greater than 6% or not greater than 5% or not greater than 4% or not greater than 3% or not greater than 2% or not greater than 1%. In an embodiment, the abrasive article can include a plurality of shaped abrasive particles having a body having at least 3 interior corners, and wherein at least 0.01% of the total number of shaped abrasive particles have a crack at an interior corner on the first major surface or at least 0.5% or at least 1% or at least 2% or at least 3% or at least 4% or at least 5% or at least 8% or at least 10%. In an embodiment, the abrasive article can include a plurality of shaped abrasive particles having a body having at least 3 interior corners, and wherein at least 0.01% of the total number of shaped abrasive particles have a crack at an interior corner on the second major surface or at least 0.5% or at least 1% or at least 2% or at least 3% or at least 4% or at least 5% or at least 8% or at least 10%. It will be appreciated that the percentage of the plurality of shaped abrasive particles having a crack at an interior corner may be between any of the minimum and maximum values noted above, including, for example, but not limited to at least 0.5% and not greater than 50% or at least 5% and not greater than 30%.

In an embodiment, the abrasive article may include a plurality of shaped abrasive particles of a particular material that may facilitate improved manufacturing or performance of the abrasive article. In an embodiment, the abrasive article may include a plurality of shaped abrasive particles including a ceramic material. In an embodiment, the abrasive article may include a plurality of shaped abrasive particles including at least one of a nitride, oxide, carbide, boride, oxynitride, oxyboride, diamond, carbon-containing material, or any combination thereof. In an embodiment, the abrasive article may include a plurality of shaped abrasive particles including an oxide compound or complex, such as aluminum oxide, zirconium oxide, titanium oxide, yttrium

oxide, chromium oxide, strontium oxide, silicon oxide, magnesium oxide, rare-earth oxides, or any combination thereof.

In an embodiment, the plurality of shaped abrasive particles can include a particular percentage of alumina that may facilitate improved performance and/or manufacturing of the abrasive article. In an embodiment, the shaped abrasive particles can include at least 80 wt. % alumina or at least 90 wt. % alumina or at least 91 wt. % alumina or at least 92 wt. % alumina or at least 93 wt. % alumina or at least 94 wt. % alumina or at least 95 wt. % alumina or at least 96 wt. % alumina or at least 97 wt. % alumina. In an embodiment, the shaped abrasive particles can include not greater than 99.5 wt. % alumina or not greater than 99 wt. % alumina or not greater than 98.5 wt. % alumina or not greater than 97.5 wt. % alumina or not greater than 97 wt. % alumina or not greater than 96 wt. % alumina or not greater than 94 wt. % alumina. It will be appreciated that the percentage of alumina in the plurality of shaped abrasive particles may be between any of the minimum and maximum values noted above, including, for example, but not limited to, at least 80 wt. % and no greater than 99 wt. % or at least 93 wt. % and no greater than 97 wt. %.

In an embodiment, each shaped abrasive particle of the plurality of shaped abrasive particles can have a particular density that may facilitate improved manufacturing and/or performance of the abrasive article. In an embodiment, each shaped abrasive particle of the plurality of shaped abrasive particles can have a density of at least 95% theoretical density.

In an embodiment, each shaped abrasive particle of the plurality of shaped abrasive particles may have a particular grain size that may facilitate improved manufacturing and/or performance of the abrasive particles. In an embodiment, each shaped abrasive particle of the plurality of shaped abrasive particles may have an average grain (crystallite) size of not greater than 1 micron or not greater than 0.8 microns or not greater than 0.6 microns or not greater than 0.4 microns or not greater than 0.2 microns as measured according to the uncorrected intercept method. In an embodiment, each shaped abrasive particle of the plurality of shaped abrasive particles may have an average grain (crystallite) size of at least 0.01 microns or at least 0.05 microns. It will be appreciated that the grain size of the plurality of shaped abrasive particles may be between any of the minimum and maximum values noted above, including, for example, but not limited to, at least 0.01 microns and no greater than 1 micron or at least 0.05 microns and no greater than 0.8 microns.

In an embodiment, the abrasive article can include a particular density of shaped abrasive particles that may facilitate improved performance and/or manufacturing of the abrasive article. In an embodiment, the density of the plurality of shaped abrasive particle per square centimeter of the abrasive article may be not greater than about 70 particles/cm² or not greater than 65 particles/cm² or not greater than 60 particles/cm² or not greater than 55 particles/cm² or not greater than about 50 particles/cm². In an embodiment, the density of the plurality of shaped abrasive particles per square centimeter of the abrasive article is at least 5 particles/cm² or at least 10 particles/cm². It will be appreciated that the density of the plurality of shaped abrasive particles may be between any of the minimum and maximum values noted above, including, for example, but not limited to, at least 5 particles/cm² and no greater than 60 particles/cm² or at least 10 particles/cm² and no greater than 50 particles/cm².

In an embodiment, the abrasive article can include a particular density of well oriented abrasive particles that may facilitate improved performance and/or manufacturing of the abrasive article. In an embodiment, the density of well oriented abrasive particle per square centimeter of the abrasive article may be at least 42 grains/cm² or at least 43 grains/cm² or at least 44 grains/cm² or at least 45 grains/cm² or at least 46 grains/cm² or at least 47 grains/cm² or at least 48 grains/cm² or at least 49 grains/cm² or at least 50 grains/cm² or at least 51 grains/cm² or at least 52 grains/cm² or at least 53 grains/cm² or at least 54 grains/cm². In an embodiment, the density of well oriented abrasive particles per square centimeter of the abrasive can be not greater than 100 grains/cm² or not greater than 95 grains/cm² or not greater than 90 grains/cm² or not greater than 85 grains/cm² or not greater than 80 grains/cm² or not greater than 75 grains/cm² or not greater than 70 grains/cm² or not greater than 65 grains/cm² or not greater than 60 grains/cm². It will be appreciated that the density of well oriented abrasive particles may be between any of the minimum and maximum values noted above, including, for example, but not limited to, at least 42 particles/cm² and no greater than 60 particles/cm² or at least 49 particles/cm² and no greater than 70 particles/cm².

In an embodiment, the abrasive article can include a particular weight of make coat that may facilitate improved performance and/or manufacturing of the abrasive article. In an embodiment, the abrasive article can include at least than 1 lbs./rm or at least 2 lbs./rm or at least 3 lbs./rm or at least 4 lbs./rm or at least 5 lbs./rm or at least 6 lbs./rm or at least 7 lbs./rm or at least 8 lbs./rm or at least 9 lbs./rm or at least 10 lbs./rm or at least 11 lbs./rm or at least 12 lbs./rm or at least 13 lbs./rm or at least 14 lbs./rm or at least 15 lbs./rm or at least 16 lbs./rm. In another embodiment, the abrasive article can include not greater than 20 lbs./rm or not greater than 19.5 lbs./rm or not greater than 19 lbs./rm or not greater than 18.5 lbs./rm or not greater than 18 lbs./rm or not greater than 17.5 lbs./rm or not greater than 17 lbs./rm. It will be appreciated that the weight of make coat can be between any of the minimum and maximum values noted above, including, for example, at least 9 lbs./rm and not greater than 20 lbs./rm or at least 12 lbs./rm and not greater than 18.5 lbs./rm.

In an embodiment, the coated abrasive article can include an abrasive surface including the abrasive particles. In an embodiment, a certain percentage of total surface area of the abrasive surface can include the plurality of shaped abrasive particles. In an embodiment, not greater than 90% of a total surface area of the abrasive surface comprises the plurality of shaped abrasive particles or not greater than 80% or not greater than 70% or not greater than 60% or not greater than 50% or not greater than 40% or not greater than 30% or not greater than 20%. In an embodiment, at least 1% of the total surface area of the abrasive surface comprises the plurality of shaped abrasive particles or at least 5% or at least 8% or at least 10% or at least 15% or at least 20% or at least 25% or at least 30% or at least 35% or at least 40% or at least 45% or at least 50%. It will be appreciated that the percentage of total surface area of the abrasive surface including the plurality of shaped abrasive particles may be between any of the minimum and maximum values noted above, including for example, but not limited to, at least 5% and no greater than 50% or at least 15% and no greater than 80%.

Many different aspects and embodiments are possible. Some of those aspects and embodiments are described herein. After reading this specification, skilled artisans will appreciate that those aspects and embodiments are only illustrative and do not limit the scope of the present inven-

tion. Embodiments may be in accordance with any one or more of the embodiments as listed below.

EMBODIMENTS

Embodiment 1. An abrasive article comprising:

a backing;

a make coat overlying the backing;

a plurality of abrasive particles overlying the backing and at least partially contained in the make coat; and

a make coat thickness ratio (T_g/T_a) of not greater than 1.5, wherein T_g is the average thickness of the make coat at the sides of the abrasive particles and the T_a is the average thickness of the make coat.

Embodiment 2. The abrasive article of embodiment 1, wherein the abrasive particles comprise shaped abrasive particles and/or elongated abrasive particles.

Embodiment 3. The abrasive article of embodiment 1, wherein the thickness ratio (T_g/T_a) is not greater than 1.45 or not greater than 1.43 or not greater than 1.40 or not greater than 1.38 or not greater than 1.35 or not greater than 1.33 or not greater than 1.30 or not greater than 1.28 or not greater than 1.25 or not greater than 1.23 or not greater than 1.20 or not greater than 1.18 or not greater than 1.15 or not greater than 1.13 or not greater than 1.10 or not greater than 1.08 or not greater than 1.05 or not greater than 1.03.

Embodiment 4. The abrasive article of embodiment 1, wherein the thickness ratio (T_g/T_a) is at least 0.70 or at least 0.80 or at least 0.90 or at least 0.98 or at least 1.00 or at least 1.03 or at least 1.05 or at least 1.08.

Embodiment 5. The abrasive article of embodiment 1, wherein the average thickness of the make coat at the sides of the grains (T_g) is at least 50 microns or at least 60 microns or at least 70 microns or at least 80 microns or at least 90 microns or at least 100 microns or at least 110 microns or at least 120 microns or at least 130 microns or at least 140 microns or at least 150 microns.

Embodiment 6. The abrasive article of embodiment 1, wherein the average thickness of the make coat at the sides of the grains (T_g) is not greater than 1 mm or not greater than 800 microns or not greater than 700 microns or not greater than 600 microns or not greater than 500 microns or not greater than 400 microns or not greater than 300 microns.

Embodiment 7. The abrasive article of embodiment 1, wherein the make coat comprises a thickness standard deviation at the sides of the abrasive particles (STD T_g) of not greater than 100 microns or not greater than 90 microns or not greater than 85 microns or not greater than 80 microns or not greater than 75 microns or not greater than 70 microns or not greater than 65 microns or not greater than 60 microns or not greater than 55 microns or not greater than 50 microns or not greater than 45 microns or not greater than 40 microns or not greater than 35 microns or not greater than 30 microns.

Embodiment 8. The abrasive article of embodiment 7, wherein the make coat comprises a thickness standard deviation at the sides of the abrasive particles (STD T_g) of at least 1 micron or at least 5 microns or at least 10 microns or at least 15 microns or at least 20 microns or at least 25 microns or at least 30 microns.

Embodiment 9. The abrasive article of embodiment 1, wherein the average thickness of the make coat (T_a) is at least 50 microns or at least 60 microns or at least 70 microns or at least 80 microns or at least 90 microns or at least 100 microns or at least 110 microns or at least 120 microns or at least 130 microns or at least 140 microns or at least 150 microns or at least 160 microns.

Embodiment 10. The abrasive article of embodiment 1, wherein the average thickness of the make coat (T_a) is not greater than 1 mm or not greater than 800 microns or not greater than 700 microns or not greater than 600 microns or not greater than 500 microns or not greater than 400 microns or not greater than 300 microns or not greater than 275 microns or not greater than 250 microns or not greater than 225 microns or not greater than 200 microns.

Embodiment 11. The abrasive article of embodiment 1, wherein the make coat comprises a thickness standard deviation (STD T) of not greater than 100 microns or not greater than 90 microns or not greater than 85 microns or not greater than 80 microns or not greater than 75 microns or not greater than 70 microns or not greater than 65 microns or not greater than 60 microns or not greater than 55 microns or not greater than 50 microns or not greater than 45 microns or not greater than 40 microns or not greater than 35 microns or not greater than 30 microns or not greater than 25 microns or not greater than 20 microns or not greater than 15 microns or not greater than 10 microns.

Embodiment 12. The abrasive article of embodiment 11, wherein the make coat comprises a thickness standard deviation (STD T) of at least 1 micron or at least 2 microns or at least 3 microns or at least 4 microns or at least 5 microns or at least 7 microns or at least 10 microns or at least 12 microns or at least 15 microns or at least 18 microns or at least 20 microns or at least 22 microns or at least 25 microns or at least 28 microns or at least 30 microns.

Embodiment 13. The abrasive article of embodiment 1, wherein at least a portion of the abrasive particles comprise a random rotational orientation.

Embodiment 14. The abrasive article of embodiment 13, wherein a portion includes at least 10% of the total number of abrasive particles or at least 20% or at least 30% or at least 40% or at least 50% or at least 60% or at least 70% or at least 80% or at least 90% or essentially all the abrasive particles have a random rotational orientation.

Embodiment 15. The abrasive article of embodiment 1, further comprising a standing portion of abrasive particles have a standing orientation, wherein the standing portion includes at least 10% of the total number of the abrasive particles or at least 20% or at least 30% or at least 40% or at least 50% or at least 55% or at least 57% or at least 60% or at least 62% or at least 65% or at least 67% or at least 70% or at least 72% or at least 75% or at least 77% or at least 80% or at least 82% or at least 85% or at least 87% or at least 90% of the total number of the abrasive particles.

Embodiment 16. The abrasive article of embodiment 15, wherein the standing portion is not greater than 99.9% of the total number of the abrasive particles or not greater than 99% or not greater than 98% or not greater than 97% or not greater than 96% or not greater than 95% of the total number of the abrasive particles.

Embodiment 17. The abrasive article of embodiment 1, further comprising a slanted portion of abrasive particles have a slanted orientation, wherein the slanted portion includes at least 1% of the total number of the abrasive particles or at least 2% or at least 3% or at least 4% or at least 5% or at least 6% or at least 7% or at least 8% or at least 9% or at least 10% or at least 11% or at least 12% or at least 13% or at least 14% or at least 15% or at least 16% or at least 17% or at least 18% or at least 20% or at least 25% of the total number of the abrasive particles.

Embodiment 18. The abrasive article of embodiment 17, wherein the slanted portion is not greater than 90% of the total number of the abrasive particles or not greater than 85% or not greater than 80% or not greater than 75% or not

greater than 70% or not greater than 65% or not greater than 60% or not greater than 55% or not greater than 50% or not greater than 45% or not greater than 40% or not greater than 35% or not greater than 30% or not greater than 25% or not greater than 20% or not greater than 18% or not greater than 15% or not greater than 12% or not greater than 10% of the total number of the abrasive particles.

Embodiment 19. The abrasive article of embodiment 1, further comprising a standing portion of abrasive particles (Pst) having a standing orientation and a slanted portion (Psl) of abrasive particles having a slanted orientation, and further comprising a ratio of the standing portion relative to the slanted portion (PSt/Psl) of at least 1 or at least 1.2 or at least 1.4 or at least 1.6 or at least 1.8 or at least 2.0 or at least 2.2 or at least 2.4 or at least 2.6 or at least 2.8 or at least 3.0 or at least 3.2 or at least 3.4 or at least 3.6 or at least 3.8 or at least 4.0 or at least 4.2 or at least 4.4 or at least 4.6 or at least 4.8 or at least 5.0 or at least 5.2 or at least 5.4 or at least 5.6.

Embodiment 20. The abrasive article of embodiment 19, wherein the ratio of the standing portion relative to the slanted portion (PS/PSl) is not greater than 100 or not greater than 95 or not greater than 90 or not greater than 80 or not greater than 70 or not greater than 60 or not greater than 50 or not greater than 40 or not greater than 30 or not greater than 20 or not greater than 10.

Embodiment 21. The abrasive article of embodiment 1, further comprising a fallen portion of abrasive particles have a fallen orientation, wherein the fallen portion includes at least 0.1% of the total number of the abrasive particles or at least 0.2% or at least 0.4% or at least 0.6% or at least 0.8% or at least 1% or at least 1.5% or at least 2% or at least 2.5% or at least 3% or at least 3.5% or at least 4% or at least 4.5% or at least 5% of the total number of the abrasive particles.

Embodiment 22. The abrasive article of embodiment 21, wherein the fallen portion is not greater than 20% of the total number of the abrasive particles or not greater than 18% or not greater than 15% or not greater than 12% or not greater than 10% or not greater than 9% or not greater than 8% or not greater than 7% or not greater than 6% or not greater than 5% of the total number of the abrasive particles.

Embodiment 23. The abrasive particle of embodiment 1, further comprising a standing portion of abrasive particles (Pst) having a standing orientation and a fallen portion (Pf) of abrasive particles having a fallen orientation, and further comprising a ratio of the standing portion relative to the fallen portion (Pst/Pf) of at least 2.0 or at least 2.2 or at least 2.4 or at least 2.6 or at least 2.8 or at least 3.0 or at least 3.2 or at least 3.4 or at least 3.6 or at least 3.8 or at least 4.0 or at least 4.2 or at least 4.4 or at least 4.6 or at least 4.8 or at least 5.0 or at least 5.2 or at least 5.4 or at least 5.6 or at least 5.8 or at least 6.0 or at least 6.2 or at least 6.4 or at least 6.6 or at least 6.8 or at least 7.0 or at least 7.2 or at least 7.4 or at least 7.6 or at least 7.8 or at least 8.0.

Embodiment 24. The abrasive article of embodiment 23, wherein the ratio of the standing portion relative to the fallen portion (PSt/Pf) is not greater than 1000 or not greater than 800 or not greater than 500 or not greater than 200 or not greater than 100 or not greater than 95.

Embodiment 25. The abrasive article of embodiment 1, further comprising a slanted portion of abrasive particles (PsI) having a slanted orientation and a fallen portion (Pt) of abrasive particles having a fallen orientation, and further comprising a ratio of the slanted portion relative to the fallen portion (PSI/Pf) of at least 1 or at least 1.2 or at least 1.4 or at least 1.6 or at least 1.8 or at least 2.0 or at least 2.2 or at least 2.4 or at least 2.6 or at least 2.8 or at least 3.0 or at least

3.2 or at least 3.4 or at least 3.6 or at least 3.8 or at least 4.0 or at least 4.2 or at least 4.4 or at least 4.6 or at least 4.8 or at least 5.0 or at least 5.2 or at least 5.4 or at least 5.6.

Embodiment 26. The abrasive article of embodiment 25, wherein the ratio of the slanted portion relative to the fallen portion (PSI/Pf) is not greater than 100 or not greater than 95 or not greater than 90 or not greater than 80 or not greater than 70 or not greater than 60 or not greater than 50 or not greater than 40 or not greater than 30 or not greater than 20 or not greater than 10 or not greater than 8 or not greater than 6.

Embodiment 27. The abrasive article of embodiment 1, further comprising an inverted portion of abrasive particles have an inverted orientation, wherein the inverted portion includes at least 0.1% of the total number of the abrasive particles or at least 0.2% or at least 0.4% or at least 0.6% or at least 0.8% or at least 1% or at least 1.5% or at least 2% or at least 2.5% or at least 3% or at least 3.5% or at least 4% or at least 4.5% or at least 5% of the total number of the abrasive particles.

Embodiment 28. The abrasive article of embodiment 27, wherein the inverted portion is not greater than 20% of the total number of the abrasive particles or not greater than 18% or not greater than 15% or not greater than 12% or not greater than 10% or not greater than 9% or not greater than 8% or not greater than 7% or not greater than 6% or not greater than 5% of the total number of the abrasive particles.

Embodiment 29. The abrasive article of embodiment 1, further comprising a standing portion of abrasive particles (Pst) having a standing orientation and an inverted portion (Pi) of abrasive particles having an inverted orientation, and further comprising a ratio of the standing portion relative to the inverted portion (PSt/Pi) of at least 1 or at least 2.0 or at least 3.0 or at least 4.0 or at least 5.0 or at least 6 or at least 7 or at least 8 or at least 9 or at least 10 or at least 12 or at least 15 or at least 18 or at least 20 or at least 25 or at least 30 or at least 40 or at least 50.

Embodiment 30. The abrasive article of embodiment 29, wherein the ratio of the standing portion to the inverted portion (PSt/Pi) of not greater than 100 or not greater than 95 or not greater than 90 or not greater than 80 or not greater than 70 or not greater than 60 or not greater than 50 or not greater than 40 or not greater than 30 or not greater than 20 or not greater than 10 or not greater than 8 or not greater than 6.

Embodiment 31. The abrasive article of embodiment 1, further comprising a slanted portion of abrasive particles (Psl) having a slanted orientation and an inverted portion (Pi) of abrasive particles having an inverted orientation, and further comprising a ratio of the slanted portion relative to the inverted portion (PSI/Pi) of at least 0.6 or at least 0.7 or at least 0.8 or at least 0.9 or at least 1 or at least 1.2 or at least 1.4 or at least 1.6 or at least 1.8 or at least 2.0 or at least 2.2 or at least 2.4 or at least 2.6 or at least 2.8 or at least 3.0 or at least 3.2 or at least at least 3.4 or at least 3.6 or at least 3.8 or at least 4.0 or at least 4.2 or at least 4.4 or at least 4.6 or at least 4.8 or at least 5.0 or at least 5.2 or at least 5.4 or at least 5.6.

Embodiment 32. The abrasive article of embodiment 30, wherein the ratio of the slanted portion relative to the inverted portion (PSI/P_i) of not greater than 100 or not greater than 95 or not greater than 90 or not greater than 80 or not greater than 70 or not greater than 60 or not greater than 50 or not greater than 40 or not greater than 30 or not greater than 20 or not greater than 10 or not greater than 8

or not greater than 6 or not greater than 4 or not greater than 3 or not greater than 2 or not greater than 1.5.

Embodiment 33. The abrasive article of embodiment 1, further comprising a fallen portion of abrasive particles (Pf) having a fallen orientation and an inverted portion (Pi) of abrasive particles having an inverted orientation, and further comprising a ratio of the fallen portion relative to the inverted portion (Pf/Pi) of at least 0.6 or at least 0.7 or at least 0.8 or at least 0.9 or at least 1 or at least 1.2 or at least 1.4 or at least 1.6 or at least 1.8 or at least 2.0 or at least 2.2 or at least 2.4 or at least 2.6 or at least 2.8 or at least 3.0 or at least 3.2 or at least 3.4 or at least 3.6 or at least 3.8 or at least 4.0 or at least 4.2 or at least 4.4 or at least 4.6 or at least 4.8 or at least 5.0.

Embodiment 34. The abrasive article of embodiment 33, wherein the ratio of the fallen portion relative to the inverted portion (Pf/Pi) of not greater than 100 or not greater than 95 or not greater than 90 or not greater than 80 or not greater than 70 or not greater than 60 or not greater than 50 or not greater than 40 or not greater than 30 or not greater than 20 or not greater than 10 or not greater than 8 or not greater than 6 or not greater than 4 or not greater than 3 or not greater than 2 or not greater than 1.5.

Embodiment 35. The abrasive article of embodiment 1, further comprising a standing portion of abrasive particles (Pst) having a standing orientation and a slanted portion (Psl) of abrasive particles having a slanted orientation, and further comprising a well-oriented percentage represented by the sum of the standing portion (%) plus the slanted portion (%) relative to all of the abrasive particles (i.e., 100%), wherein the well-oriented percentage is at least 60% or at least 62% or at least 65% or at least 67% or at least 70% or at least 72% or at least 75% or at least 77% or at least 80% or at least 82% or at least 85% or at least 87% or at least 90% or at least 92% or at least 95%.

Embodiment 36. The abrasive article of embodiment 35, wherein the well-oriented percentage is not greater than 99.9% or not greater than 99% or not greater than 98% or not greater than 97% or not greater than 96% or not greater than 95%.

Embodiment 37. The abrasive article of embodiment 1, further comprising a coating density of well oriented particles of at least 42 grains/cm² or at least 43 grains/cm² or at least 44 grains/cm² or at least 45 grains/cm² or at least 46 grains/cm² or at least 47 grains/cm² or at least 48 grains/cm² or at least 49 grains/cm² or at least 50 grains/cm² or at least 51 grains/cm² or at least 52 grains/cm² or at least 53 grains/cm² or at least 54 grains/cm².

Embodiment 38. The abrasive article of embodiment 37, comprising a coating density of well oriented particles of not greater than 100 grains/cm² or not greater than 95 grains/cm² or not greater than 90 grains/cm² or not greater than 85 grains/cm² or not greater than 80 grains/cm² or not greater than 75 grains/cm² or not greater than 70 grains/cm² or not greater than 65 grains/cm² or not greater than 60 grains/cm².

Embodiment 39. The abrasive article of embodiment 1, wherein the make coat comprises a make coat add on weight of not greater than 20 lbs./rm or not greater than 19.5 lbs./rm or not greater than 19 lbs./rm or not greater than 18.5 lbs./rm or not greater than 18 lbs./rm or not greater than 17.5 lbs./rm or not greater than 17 lbs./rm.

Embodiment 40. The abrasive article of embodiment 39, wherein the make coat comprises a make coat add on weight of at least 1 lbs./rm or at least 2 lbs./rm or at least 3 lbs./rm or at least 4 lbs./rm or at least 5 lbs./rm or at least 6 lbs./rm or at least 7 lbs./rm or at least 8 lbs./rm or at least 9 lbs./rm or at least 10 lbs./rm or at least 11 lbs./rm or at least

12 lbs./rm or at least 13 lbs./rm or at least 14 lbs./rm or at least 15 lbs./rm or at least 16 lbs./rm.

Embodiment 41. The abrasive article of embodiment 1, wherein the abrasive particles include shaped abrasive particles or elongated abrasive particles, and where each of the shaped abrasive particles or elongated abrasive particles include a body having a length (l), a width (w) and a thickness (t), wherein the width > thickness and the length > thickness.

Embodiment 42. The abrasive article of embodiment 41, wherein the shaped abrasive particles having a 3-PT star two-dimensional shape as viewed in a plane of a length and width of the body.

Embodiment 43. The abrasive article of embodiment 41, wherein each of the shaped abrasive particles or elongated abrasive particles of the plurality of shaped abrasive particles or elongated abrasive particles comprises a body including a first major surface, a second a major surface opposite the first major surface, and a side surface extending between the first major surface and the second major surface, wherein the body comprises at least 3 exterior corners and wherein the side surface comprises at least 4 side surface sections, or at least 5 side surface sections or at least 6 side surface sections.

Embodiment 44. The abrasive article of embodiment 41, wherein the plurality of shaped abrasive particles or elongated abrasive particles comprises an average side surface angle between the side surface and the first major surface of at least 70 degrees and not greater than 94 degrees or within a range of at least 80 degrees and not greater than 93 degrees or within a range of at least 83 degrees and not greater than 92 degrees or within a range of at least 85 degrees and not greater than 91 degrees.

Embodiment 45. The abrasive article of embodiment 41, wherein the plurality of shaped abrasive particles or elongated abrasive particles comprise an average side surface angle between the side surface and the second major surface of at least 70 degrees and not greater than 94 degrees or within a range of at least 80 degrees and not greater than 93 degrees or within a range of at least 83 degrees and not greater than 92 degrees or within a range of at least 85 degrees and not greater than 91 degrees.

Embodiment 46. The abrasive article of embodiment 41, wherein each of the shaped abrasive particles or elongated abrasive particles of the plurality of shaped abrasive particles or elongated abrasive particles comprises a body having at least 3 exterior corners, wherein the sum of the angles of the exterior corners is less than 180 degrees.

Embodiment 47. The abrasive article of embodiment 46, wherein each of the shaped abrasive particles or elongated abrasive particles of the plurality of shaped abrasive particles or elongated abrasive particles comprises a body having at least 3 exterior corners, wherein each of the exterior corners defines an angle less than 60 degrees or less than 59 degrees or less than 58 degrees or less than 57 degrees or less than 56 degrees or less than 55 degrees.

Embodiment 48. The abrasive article of embodiment 41, wherein each of the shaped abrasive particles or elongated abrasive particles of the plurality of shaped abrasive particles or elongated abrasive particles comprises a body having at least 3 exterior corners and at least 3 interior corners, wherein each of the interior corners have an interior corner angle value greater than any of the exterior corner values of any of the at least 3 exterior corners.

Embodiment 49. The abrasive article of embodiment 41, wherein each of the shaped abrasive particles or elongated abrasive particles of the plurality of shaped abrasive particles

ticles or elongated abrasive particles comprises a body having at least 3 interior corners, and wherein not greater than 50% of the total number of a plurality of shaped abrasive particles or elongated abrasive particles have a crack at an interior corner or not greater than 45% or not greater than 40% or not greater than 35% or not greater than 30% or not greater than 25% or not greater than 20% or not greater than 18% or not greater than 16% or not greater than 14% or not greater than 12% or not greater than 10% or not greater than 9% or not greater than 8% or not greater than 7% or not greater than 6% or not greater than 5% or not greater than 4% or not greater than 3% or not greater than 2% or not greater than 1%.

Embodiment 50. The abrasive article of embodiment 49, wherein each of the shaped abrasive particles or elongated abrasive particles of the plurality of shaped abrasive particles or elongated abrasive particles comprises a body having at least 3 interior corners, and wherein at least 0.01% of the total number of a plurality of shaped abrasive particles or elongated abrasive particles have a crack at an interior corner or at least 0.5% or at least 1% or at least 2% or at least 3% or at least 4% or at least 5% or at least 8% or at least 10%.

Embodiment 51. The abrasive article of embodiment 41, wherein the abrasive particles comprise a ceramic material.

Embodiment 52. The abrasive article of embodiment 51, wherein the abrasive particles comprise at least one of a nitride, oxide, carbide, boride, oxynitride, oxyboride, diamond, carbon-containing material, or any combination thereof.

Embodiment 53. The abrasive article of embodiment 51, wherein the abrasive particles comprise an oxide compound or complex, such as aluminum oxide, zirconium oxide, titanium oxide, yttrium oxide, chromium oxide, strontium oxide, silicon oxide, magnesium oxide, rare-earth oxides, or any combination thereof.

Embodiment 54. The abrasive article of embodiment 51, wherein the abrasive particles comprise at least 80 wt. % alumina or at least 90 wt. % alumina or at least 91 wt. % alumina or at least 92 wt. % alumina or at least 93 wt. % alumina or at least 94 wt. % alumina or at least 95 wt. % alumina or at least 96 wt. % alumina or at least 97 wt. % alumina.

Embodiment 55. The abrasive article of embodiment 51, wherein the abrasive particles comprise not greater than 99.5 wt. % alumina or not greater than 99 wt. % alumina or not greater than 98.5 wt. % alumina or not greater than 97.5 wt. % alumina or not greater than 97 wt. % alumina or not greater than 96 wt. % alumina or not greater than 94 wt. % alumina.

Embodiment 56. The abrasive article of embodiment 1, wherein the abrasive particles have an average density of at least 95% theoretical density.

Embodiment 57. The abrasive article of embodiment 1, wherein the abrasive particles comprise an average grain (crystallite) size of not greater than 1 micron or not greater than 0.8 microns or not greater than 0.6 microns or not greater than 0.4 microns or not greater than 0.2 microns as measured according to the uncorrected intercept method.

Embodiment 58. The abrasive article of embodiment 57, wherein each shaped abrasive particles or elongated abrasive particles of the plurality of shaped abrasive particles or elongated abrasive particles comprises an average grain (crystallite) size of at least 0.01 microns or at least 0.05 microns.

Embodiment 59. The abrasive article of embodiment 1, wherein an areal density of the abrasive particles per square centimeter of the abrasive article may be not greater than

about 70 particles/cm² or not greater than 65 particles/cm² or not greater than 60 particles/cm² or not greater than 55 particles/cm² or not greater than about 50 particles/cm².

Embodiment 60. The abrasive article of embodiment 59, wherein the areal density is at least 5 particles/cm² or at least 10 particles/cm².

Embodiment 61. The abrasive article of embodiment 1, further comprising an abrasive surface including the abrasive particles and at least one adhesive layer, wherein not greater than 90% of a total surface area of the abrasive surface includes the abrasive particles or not greater than 80% or not greater than 70% or not greater than 60% or not greater than 50% or not greater than 40% or not greater than 30% or not greater than 20%.

Embodiment 62. The abrasive article of embodiment 1, wherein at least 1% of the total surface area of the abrasive surface comprises the abrasive particles or at least 5% or at least 8% or at least 10% or at least 15% or at least 20% or at least 25% or at least 30% or at least 35% or at least 40% or at least 45% or at least 50%.

Embodiment 63. The abrasive article of embodiment 1, wherein the abrasive particles include a first group of abrasive particles and a second group of abrasive particles.

Embodiment 64. The abrasive article of embodiment 63, wherein the first group of abrasive particles includes at least two different types of shaped abrasive particles, wherein the two different types of shaped abrasive particles are different from each other based on at least one characteristic selected from the group of particle size, two-dimensional shape, three-dimensional shape, composition, hardness, toughness, friability, density, grain size, agglomeration state, lateral position, longitudinal position, rotational orientation, or any combination thereof.

Embodiment 65. The abrasive article of embodiment 63, further comprising a second group of abrasive particles different than the first group of abrasive particles.

Embodiment 66. The abrasive article of embodiment 65, wherein the second group of abrasive particles comprises diluent abrasive particles.

Embodiment 67. The abrasive article of embodiment 65, wherein the second group of abrasive particles comprises randomly shaped abrasive particles.

Embodiment 68. The abrasive article of embodiment 65, wherein the second group of abrasive particles are different from the first group of abrasive particles based on at least one characteristic selected from the group of particle size, two-dimensional shape, three-dimensional shape, composition, hardness, toughness, friability, density, grain size, agglomeration state, lateral position, longitudinal position, rotational orientation, or any combination thereof.

Embodiment 69. A coated abrasive article having the features of embodiment 1, including the backing having a major surface and an abrasive layer forming an abrasive surface overlying the major surface of the backing, wherein the abrasive layer forms substantially a single layer of abrasive particles adhered to the major surface of the backing by one or more adhesive layers.

Embodiment 70. An abrasive article comprising:

- a backing;
- a make coat overlying the backing;
- a plurality of abrasive particles overlying the backing and at least partially contained in the make coat; and
- a thickness standard deviation of the make coat at the sides of the plurality of abrasive particles (STDTg) of at least 1 micron or not greater than 60 microns.

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Embodiment 71. The abrasive article of embodiment 66, further comprising any one or a combination of any of the claims or embodiments herein.

Embodiment 72. A process for forming a coated abrasive article comprising:

providing a backing;

forming a make coat overlying the backing;

placing a plurality of abrasive particles overlying the backing and at least partially contained in the make coat; and

a thickness standard deviation of the make coat at the sides of the plurality of abrasive particles (STDTg) of at least 1 micron or not greater than 60 microns.

Embodiment 73. The abrasive article of embodiment 1, wherein the make coat comprises wollastonite, PF resin, water, or a combination thereof.

Embodiment 74. The abrasive article of embodiment 1, wherein the make coat comprises a viscosity of at least 3500 cps or at least 3750 cps or at least 4000 cps or at least 4250 cps or at least 4500 cps or at least 4750 cps or at least 5000 cps or at least 5250 cps.

Embodiment 75. The abrasive article of embodiment 1, wherein the make coat comprises a viscosity of no greater than 7000 cps or no greater than 6750 cps or no greater than 6500 cps or no greater than 6250 cps or no greater than 6000 cps or no greater than 5750.

Embodiment 76. The process of embodiment 72 wherein the abrasive article is the abrasive article of embodiment 1+70 or 73-75.

EXAMPLES

Example 1

Sample 1 and Comparative Sample 1 were prepared according to the following procedure having the specifications outlined in table 1. Backings were saturated with 20.5 lbs./rm of the following composition:

Latex: 63.85%

Cab-o-sil: 0.98%

Defoamer: 0.44%

Wetting Agent: 0.24%

Calcium Carbonate: 31.93%

Dye (reddish/pink): 2.56%

The saturated backing was backfilled with 7.1 lbs./rm of the following composition:

PF Resin: 59.13%

Defoamer: 0.3%

Wetting Agent: 0.66%

Solmod Tamol 165A: 2.01%

Wollastonite: 19.71%

Red Dye: 0.21%

Water: 17.98%

A make coat is applied to the saturated and backfilled backing via two roll coating. The make coat thickness is controlled by nip gap to achieve the desired add on weight. Abrasive particles are then applied to the wet make and the backing via electrostatic coating. The backing, make, and grains are then cured in an oven according to the curing schedule in table 1. Size and supersize coats are applied and cured in the same manner as the make coat according to the specifications in Table 1.

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TABLE 1

	Sample 1	Comparative Sample 1
5	Backing	1 ply woven PET
	Make Coat	1 ply woven PET
	PF resin	49.34 wt. %
	Silane A1100	0.44 wt. %
	Wetting Agent	0.15 wt. %
10	Wollastonite	49.34 wt. %
	Water	0.75 wt. %
	Black dye	1 wt. %
	Nip Gap	0.41 in.
	Add on Weight	16 lbs./rm
	Curing Cycle	20 min at 170° C.
15		20 min at 190° C.
		20 min at 210° C.
		20 min at 235° C.
	Abrasive Particles	
	Type	3-PT star, alumina
	Size	36 grit
20	Add on Weight	33 lbs./rm
	Size Coat	
	PF resin	46.95 wt. %
	PET-3MP PolyThiol (PTM)	4.69 wt. %
	Defoamer	0.09 wt. %
25	Solmod Tamol 165A	2.35 wt. %
	Water	3.54 wt. %
	Cryolite	41.31 wt. %
	Add on Weight	32 lbs./rm
	Curing Cycle	20 min at 180° C.
30		20 min at 200° C.
		20 min at 220° C.
		20 min at 235° C.
	Supersize Coat	
	PF resin	23 wt. %
	Defoamer	0.11 wt. %
35	Solmod Daxad 11	1.69 wt. %
	Water	7.77 wt. %
	Orange Pigment	2.78 wt. %
	KBF ₄	64.48 wt. %
	Cab-o-sil	0.17 wt. %
	Add on Weight	30 lbs./rm
40	Curing Cycle	20 min at 170° C.
		20 min at 190° C.
		20 min at 210° C.
		20 min at 235° C.

Conventional sample 1 was a 3M™ Cubitron™ II Cloth Belt 984F 36+grit.

Average make coat thickness was measured according to the following procedure. The samples were cut through the middle to reveal a cross section. The samples are then cut into 2-inch segments and mounted on an epoxy puck. Two 2-inch segments are then imaged, and the make layer is identified by coloring in the layer using the imaging software. FIG. 10 includes an example image of an abrasive article include a colored make layer. Image analysis is used to overlay vertical gridlines, and the line segments overlapping the make layer were identified and isolated. Each line segment corresponds to a make coat thickness measurement. The average of all segments was taken. Approximately 150-200 overlapping line segments were made per two-inch sample segment, resulting in over 300 measurements for each sample.

Average make coat thickness near standing grains was measured according to the following procedure. The same cross-sectional images for average make coat thickness were also used for average make coat thickness near standing grains. Only standing grains showing their cross-sectional rectangular area with their short side in contact with the

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make coat were considered. For example, in FIG. 1, grain 102 would be considered but grain 103 would not. Additionally, only isolated grains were considered. Standing grains in contact with another grain were not considered for average make coat thickness near standing grains measurements. Measurements were made from the highest point of make contacting the grain side down to the lowest point of make contacting the backing on both sides of grain. The line of measurement is made perpendicular to the backing plane. The results for average make coat thickness and average make coat thickness near standing grains measurements can be found below in Table 2.

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ash were then transferred to a 250 ml Erlenmeyer flask. 50% hydrofluoric acid was added to the flask so that the abrasive particles were just covered. The particles were allowed to soak for 30 minutes. The acid was then diluted with DI water and the liquid was disposed of. The flask was allowed to dry for 2 hours at 60-70° C. The flask and contents were then cooled to room temperature and the reclaimed and acid washed grains were removed.

Conventional sample 2 was prepared by removing the size and supersize coats from Conventional sample 1 via sand blasting.

TABLE 3

	Sample 2	Sample 3	Comparative sample 2
Backing	1 ply woven PET	1 ply woven PET	1 ply woven PET
	Make Coat		
PF resin	49.34 wt. %	49.34 wt. %	49.34 wt. %
Silane A1100	0.44 wt. %	0.44 wt. %	0.44 wt. %
Wetting Agent	0.15 wt. %	0.15 wt. %	0.15 wt. %
Wollastoinite	49.34 wt. %	49.34 wt. %	49.34 wt. %
Water	0.75 wt. %	0.75 wt. %	0.75 wt. %
Black dye	1 wt. %	1 wt. %	1 wt. %
Nip Gap	0.41 in.	0.41 in.	0.41 in.
Add on Weight	16 lbs./rm	16 lbs./rm	20 lbs./rm
Curing Cycle	20 min at 170° C.	20 min at 170° C.	20 min at 170° C.
	20 min at 190° C.	20 min at 190° C.	20 min at 190° C.
	20 min at 210° C.	20 min at 210° C.	20 min at 210° C.
	20 min at 235° C.	20 min at 235° C.	20 min at 235° C.
	Abrasive Particles		
Type	36 grit 3-PT star, alumina	Reclaimed Cubitron	Reclaimed Cubitron
Add on Weight	33 lbs./rm	43 lbs./rm	44 lbs./rm

TABLE 2

	S1	Comparative S1	Conventional S1
Ta	155 μm	157 μm	128 μm
Std Dev of Ta	84 μm	86 μm	75 μm
Ta 95% confidence interval	146-163 μm	148-166 μm	124-132 μm
Tg	186 μm	265 μm	262 μm
Std Dev of Tg	59 μm	76 μm	99 μm
Tg 95% confidence interval	167-206 μm	233-297 μm	225-299 μm
Tg/Grain height	0.16	0.22	0.22

Example 2

Samples 2 and 3 and comparative sample 2 were prepared according to the method for Sample 1 above and the specifications in table 3. No size or supersize coats were applied. Sample 2 is essentially the same as sample 1 without a size or supersize coat.

Reclaimed Cubitron grains were reclaimed according to the following procedure. A 3M™ Cubitron™ II Cloth Belt 984F 36+grit was obtained. The belt was burned until only abrasive particles and ash remained. The abrasive particles and burnoff were then allowed to soak in a 400 ml glass beaker filled with hydrochloric acid. Enough acid to just cover the grains burnoff was used. The contents were boiled for 10 minutes and then allowed to cool. The solution was then diluted with DI water and then the liquid was disposed of. The beaker with the abrasive particles was then allowed to dry for 2 hours at 60-70° C. The particles and remaining

The orientation of the particles in samples 2 and 3, comparative samples 2, and conventional sample 2 were measured according to the following procedure.

Images of each sample were taken using a z-stacking microscope. An exemplary image can be found in FIG. 9A. The image of the sample includes a visible make coat 901 and abrasive grains, e.g., 902, 903, 904, 905. ImageJ software was used to threshold and identify the grains, e.g., 902, 903, 904, 905. When necessary, overlapping grains were segmented manually. An exemplary image of the abrasive edited to identify the grains can be found in FIG. 9B. Grains were color-coded and counted by orientation based on the criteria below in table 2. An exemplary image with color coated grains can be found in FIG. 9C. Particle 902 is in a standing orientation; particle 903 is in a slanted orientation; particle 904 is in a fallen orientation; and particle 905 is in an inverted orientation. Grain orientation data can be found below in table 3. Images included a 3.2 cm² surface area of each sample.

TABLE 4

	Tilt Angle	Shape
Standing	65-90°	Visually rectangular from top-down view
Slanted	5-65°	Visually an isosceles 3-PT star from top-down view
Fallen	0-5°	Visually an equilateral 3-PT star from top-down view
Inverse	N/A	Only the tip of the grain is in contact with the make coat

TABLE 5

	Standing	Slanted	Well oriented	Fallen	Inverse	Coating density	Well Oriented Coating Density
Sample 2	73%	12%	85%	6%	10%	65 #/cm ²	57 #/cm ²
Sample 3	44%	42%	86%	10%	4%	66 #/cm ²	55 #/cm ²
Conventional	35%	23%	58%	40%	2%	63 #/cm ²	37 #/cm ²
Sample 2 Comparative	42%	17%	59%	40%	1%	70 #/cm ²	41 #/cm ²
Sample 2							

As can be seen in Table 5, samples 2 and 3 had a larger percentage of particles in desirable orientations as compared to the comparative and conventional samples. Sample 2 included a significantly larger number of standing particles as compared to all other samples.

The present application represents a departure from the state of the art. While certain publications have disclosed that it is desirable to orient shaped abrasive particles in certain orientations these publications have not enabled the degree of orientation as disclosed in the present application. Notably, it is apparent that conventional coated abrasives have a significant portion of abrasive particles placed in undesirable orientations. The industry continues to desire an enabled system and method for achieving a greater degree of control of orientation of abrasive particles in coated abrasives. The system and methods disclosed herein enable the formation of a coated abrasive articles having greater control over the orientation of particles on a backing for creation of coated abrasive articles. Moreover, the systems and methods herein may facilitate improved fine-tuned control over certain orientations, such as control over standing, slanted, fallen, and inverse orientations of grains.

The above-disclosed subject matter is to be considered illustrative, and not restrictive, and the appended claims are intended to cover all such modifications, enhancements, and other embodiments, which fall within the true scope of the present invention. Thus, to the maximum extent allowed by law, the scope of the present invention is to be determined by the broadest permissible interpretation of the following claims and their equivalents and shall not be restricted or limited by the foregoing detailed description.

The Abstract of the Disclosure is provided to comply with Patent Law and is submitted with the understanding that it will not be used to interpret or limit the scope or meaning of the claims. In addition, in the foregoing Detailed Description, various features may be grouped together or described in a single embodiment for the purpose of streamlining the disclosure.

This disclosure is not to be interpreted as reflecting an intention that the claimed embodiments require more features than are expressly recited in each claim. Rather, as the following claims reflect, inventive subject matter may be directed to less than all features of any of the disclosed embodiments. Thus, the following claims are incorporated into the Detailed Description, with each claim standing on its own as defining separately claimed subject matter.

What is claimed is:

1. An abrasive article comprising:

a backing;

a make coat overlying the backing;

a plurality of abrasive particles overlying the backing and at least partially contained in the make coat;

a make coat thickness ratio (Tg/Ta) of not greater than 1.5, wherein Tg is the average thickness of the make coat at

the sides of the abrasive particles and the Ta is the average thickness of the make coat;

and wherein at least 60% of the total number of the abrasive particles are well oriented having a tilt angle of 5 to 90 degrees; and

wherein the plurality of abrasive particles comprises a plurality of shaped abrasive particles, each of the shaped abrasive particles of the plurality of shaped abrasive particles including at least 3 interior corners, wherein each of the shaped abrasive particles comprises a body including a first major surface, a second major surface opposite the first major surface, and a side surface extending between the first major surface and the second major surface;

and wherein not greater than 50% of the total number of shaped abrasive particles have a crack at an interior corner on the first major surface.

2. The abrasive article of claim 1, wherein the body of each of the shaped abrasive particles comprises at least 3 exterior corners and wherein the side surface comprises at least 4 side surface sections.

3. The abrasive article of claim 1, wherein the shaped abrasive particles have a three-pointed star two-dimensional shape as viewed in a plane of a length and width of the body.

4. The abrasive article of claim 1, wherein the thickness ratio (Tg/Ta) is not greater than 1.5 and at least 0.70.

5. The abrasive article of claim 4, wherein the thickness ratio (Tg/Ta) is not greater than 1.5 and at least 0.80.

6. The abrasive article of claim 5, wherein the thickness ratio (Tg/Ta) is not greater than 1.5 and at least 0.90.

7. The abrasive article of claim 4, wherein the make coat comprises a thickness standard deviation at the sides of the abrasive particles (STDTg) of not greater than 70 microns.

8. The abrasive article of claim 7, wherein the make coat thickness standard deviation at the sides of the abrasive particles (STDTg) is at least 10 microns and not greater than 70 microns.

9. The abrasive article of claim 8, wherein the make coat thickness standard deviation at the sides of the abrasive particles (STDTg) is at least 10 microns and not greater than 65 microns.

10. The abrasive article of claim 7, wherein at least 70% of the total plurality of abrasive particles comprising a plurality of shaped abrasive particles are well oriented.

11. The abrasive article of claim 10, wherein at least 80% of the plurality of abrasive particles comprising a plurality of shaped abrasive particles are well oriented.

12. The abrasive article of claim 7, further comprising a coating density of well oriented particles of at least 42 grains/cm².

13. The abrasive article of claim 12, further comprising a coating density of well oriented particles of at least 46 grains/cm².

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14. The abrasive article of claim 4, further comprising a standing portion of shaped abrasive particles (Pst) having a standing orientation and a slanted portion (Psl) of shaped abrasive particles having a slanted orientation, and further comprising a ratio of the standing portion relative to the slanted portion (Pst/Psl) of at least 1 and not greater than 100.

15. The abrasive article of claim 4, further comprising a standing portion of shaped abrasive particles (Pst) having a standing orientation and a fallen portion (Pf) of shaped abrasive particles having a fallen orientation, and further comprising a ratio of the standing portion relative to the fallen portion (Pst/Pf) of at least 2.0 and not greater than 1000.

16. The abrasive article of claim 15, further comprising a percentage of fallen shaped abrasive particles having a tilt angle of 0 to 5 degrees of not greater than 20% of the plurality of abrasive particles comprising a plurality of shaped abrasive particles.

17. An abrasive article comprising:
a backing;
a make coat overlying the backing;
a plurality of abrasive particles overlying the backing and at least partially contained in the make coat;
a make coat thickness ratio (Tg/Ta) of at least 0.7 and not greater than 1.5, wherein Tg is the average thickness of the make coat at the sides of the abrasive particles and the Ta is the average thickness of the make coat;
wherein at least 60% of the total number of the abrasive particles are well oriented having a tilt angle of 5 to 90 degrees; and

wherein the make coat comprises a thickness standard deviation at the sides of the abrasive particles (STD_{Tg}) of at least 10 microns and not greater than 70 microns.

18. The abrasive article of claim 17, wherein the thickness ratio (Tg/Ta) is not greater than 1.5 and at least 1.

19. The abrasive article of claim 18, wherein at least 80% of the total number of the abrasive particles are well oriented having a tilt angle of 5 to 90 degrees.

20. The abrasive article of claim 19, wherein the shaped abrasive particles have a three-pointed star two-dimensional shape as viewed in a plane of a length and width of the body.

21. An abrasive article comprising:
a backing;
a make coat overlying the backing;
a plurality of abrasive particles overlying the backing and at least partially contained in the make coat;

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a make coat thickness ratio (Tg/Ta) of at least 0.7 and not greater than 1.5, wherein Tg is the average thickness of the make coat at the sides of the abrasive particles and the Ta is the average thickness of the make coat;

5 wherein at least 60% of the total number of the abrasive particles are well oriented having a tilt angle of 5 to 90 degrees; and

further comprising a standing portion of abrasive particles (Pst) having a standing orientation and a slanted portion (Psl) of abrasive particles having a slanted orientation, and further comprising a ratio of the standing portion relative to the slanted portion (Pst/Psl) of at least 1 and not greater than 100.

22. The abrasive article of claim 21, wherein the thickness ratio (Tg/Ta) is not greater than 1.5 and at least 1.

23. The abrasive article of claim 22, wherein at least 80% of the total number of the abrasive particles are well oriented having a tilt angle of 5 to 90 degrees.

24. The abrasive article of claim 23, wherein the shaped abrasive particles have a three-pointed star two-dimensional shape as viewed in a plane of a length and width of the body.

25. An abrasive article comprising:
a backing;
a make coat overlying the backing;
a plurality of abrasive particles overlying the backing and at least partially contained in the make coat;
a make coat thickness ratio (Tg/Ta) of at least 0.7 and not greater than 1.5, wherein Tg is the average thickness of the make coat at the sides of the abrasive particles and the Ta is the average thickness of the make coat;

30 wherein at least 60% of the total number of the abrasive particles are well oriented having a tilt angle of 5 to 90 degrees; and

further comprising a standing portion of abrasive particles (Pst) having a standing orientation and a fallen portion (Pf) of abrasive particles having a fallen orientation, and further comprising a ratio of the standing portion relative to the fallen portion (Pst/Pf) of at least 2.0 and not greater than 1000.

26. The abrasive article of claim 25, wherein the thickness ratio (Tg/Ta) is not greater than 1.5 and at least 1.

27. The abrasive article of claim 26, wherein at least 80% of the total number of the abrasive particles are well oriented having a tilt angle of 5 to 90 degrees.

28. The abrasive article of claim 27, wherein the shaped abrasive particles have a three-pointed star two-dimensional shape as viewed in a plane of a length and width of the body.

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